Differentiable transmon design with a tensor-native finite-element electromagnetics solver: gradient-based optimization of charging energy, cross-validated against Palace

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Abstract

Superconducting-qubit electromagnetic design is an iterative guess-and-check loop: the workflow's production field solvers expose no derivatives of design figures-of-merit, so state-of-practice optimizers iterate time-consuming electromagnetic simulations and inject parameter updates from separate analytic physics models (Eriksson et al., 2025). We demonstrate the missing capability — *solver-derived* design gradients from a 3D finite-element electromagnetics solve — and use it for gradient-based optimization of a transmon's charging energy E_C. GEODE-FEM is a Rust solver built on the Burn tensor framework; a discrete-adjoint layer bridges the one stage its autodiff tape cannot reach (the sparse factorization), yielding a validated 2×2 sensitivity matrix — material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$) derivatives on both the electrostatic and the driven $H(\mathrm{curl})$ operators — each finite-difference-validated with mutation tests (relative errors 10^{-9}–10^{-5}). Because the capacitance $C = \phi^{\mathsf{T}} K \phi$ is variationally stationary, its shape derivative collapses to a Hellmann–Feynman-like explicit term; composing it with the capacitance-to-E_C chain, Newton iteration drives a capacitor geometry to a target E_C, confirmed by an independent fresh solve to 1.4×10^{-15}. On the real DeviceLayout transmon, a finite-difference-validated quadratic-element shape gradient on an adaptively refined mesh drives the island capacitance C_Σ to the 93.954 fF design anchor in four accepted steps. The anchor is the published 4.14 GHz qubit frequency read as the harmonic frequency of Palace's linear eigenproblem; it replaces an 89.9 fF anchor that was mis-derived from the anharmonic formula, and the runs aimed at that value are kept as superseded. That C_Σ is the scalar-permittivity electrostatic value, with the junction's 5.5 fF self-capacitance excluded. Its gradient is a hand-derived equilibrium-energy shape derivative, solved by conjugate gradients with an algebraic-multigrid preconditioner in native f64; it does not run on the Burn tape and uses no factorization. Seven held-out refinement rounds and a certified equilibrated-flux lower bound then bracket the converged capacitance of that design in $[92.75, 93.00]$ fF, and of a resolution-corrected second design in $[93.89, 94.14]$ fF: two-sided brackets 0.25 fF wide, which decide design against target. The optimizer reaches whatever target it is given; what remains uncertain is the target. Removing the measured qubit–resonator hybridization and a series loop inductance of the junction's current path moves the target from the dressed 93.954 fF to a bare-qubit 93.853 fF and then to 93.143 fF. Both corrections are *estimates*: the coupling and the loop inductance are extrapolated, not mesh-converged, and are assumed to describe DeviceLayout's own run. Neither design brackets the most corrected target, and re-optimizing at it is future work. The convergence study is a result in its own right. Under uniform refinement the scalar-permittivity capacitance of the unmodified device falls from 137.7 fF on the committed 133k-tetrahedron mesh to below 83.02 fF, and on adapted meshes it is

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certified in $[77.40, 77.61]$ fF. The gap to the anchor that earlier versions called a model difference was therefore discretization error, and the converged capacitance of the committed geometry lies below the anchor, not above it: by at least 17.3% with scalar permittivity, and by at least 4.7% with the sapphire tensor that the lumped-element chain uses (converged below 89.50 fF). Earlier coarse-mesh optimizations lowered C_Σ to reach an anchor; they moved the wrong way and are kept as a cautionary precursor. Eigenmode/EPR derivatives remain roadmap. The correctness credential: from the identical input mesh, GEODE-FEM reproduces Palace's six eigenmodes above 4.5 GHz to $\leq 0.033\%$ (worst case 0.032%), and the transmon qubit mode at 3.45 GHz to 5×10^{-5} %. Earlier versions of this paper called that mode spurious and the qubit mode absent; a Palace run with its search target below the mode corrects both statements. The 17.49 GHz mode both solvers share is an artifact of the distributed port term, and an opt-in rank-1 port-voltage shunt removes it. The contribution is a capability, not a speed race — on raw performance the measured record shows no corner where GEODE-FEM beats Palace at scale, and we report it honestly. Every result traces to a committed artifact; statements that rest on a scratch probe, on a reading of Palace's source, or on inference are labelled as such.

1 Introduction

The hard problem in superconducting-qubit electromagnetic design is the *inverse map*: given target Hamiltonian parameters — a charging energy E_C, an anharmonicity α, coupling capacitances — produce a physical layout. Today that map is walked by hand: parameterize a geometry, run a 3D electromagnetic simulation, extract the parameters, compare, adjust, repeat. The state-of-practice automated workflow makes the loop's structure explicit: QDesignOptimizer (Eriksson et al., 2025) iterates time-consuming electromagnetic simulations of HFSS-class solvers and guides the parameter updates with *separate*, user-defined analytic physics models — precisely because the field solver at the center of the loop produces observables but no derivatives of those observables with respect to the design. The gradient the optimizer actually wants, $\partial(\text{figure of merit})/\partial(\text{geometry, materials})$ through the field solve itself, is structurally unavailable from the production solvers — commercial (HFSS, COMSOL) and open-source (Palace (Palace developers, AWS Center for Quantum Computing, 2023)) alike: their assembly and factorization pipelines are hand-written kernels with no derivative path, and none exposes an adjoint of its solve.

The surrounding literature brackets this gap without closing it. At the *lumped-circuit* level, differentiation is a solved problem: SQcircuit (Rajabzadeh et al., 2023) analyzes arbitrary superconducting circuits, and its gradient-based follow-up (Rajabzadeh et al., 2024) computes eigenvalue and eigenvector gradients of the circuit Hamiltonian via PyTorch autodiff, optimizing qubit designs in circuit-parameter space. But the map from geometry and materials *to* those lumped parameters still runs through non-differentiable 3D electromagnetic simulation. At the *field* level, differentiable electromagnetics is owned by photonics: adjoint and autodiff methods on FDTD and integral-equation solvers are a mature inverse-design industry (Hammond et al., 2022; Hughes et al., 2019; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022), and differentiable FEM at scale has been proven outside electromagnetics by JAX-FEM in solid mechanics (Xue et al., 2023). What none of these supplies is the regime superconducting-qubit (and, more broadly, RF) design lives in: *frequency-domain finite elements on body-fitted tetrahedra with anisotropic materials* — distributed, field-level EM gradients from the same FEM solve that produces the design observables. That wedge is, to our knowledge, uncontested; this paper occupies it.

GEODE-FEM ("geode-fem") is positioned to close the gap by construction rather than retrofit. It is a full-wave 3D $H(\text{curl})$/Nédélec (plus scalar electrostatic) finite-element solver written in Rust on the Burn tensor framework (Tracel AI, 2024): element assembly is batched $[n_{\text{elem}}, 6, 6]$ tensor algebra on an autodiff-capable substrate, so the derivative of assembly with respect to materials and node coordinates is available where a classical solver has only hand-written kernels. One stage

breaks the tape: the sparse direct factorization (faer (Kazdadi, 2026)) at the heart of every solve. A discrete-adjoint layer closes exactly that gap — one extra (transpose-)solve, reusing the forward factorization, restores the full gradient (Section 8). The resulting capability is general — shape and material sensitivities for RF, photonic, and electrostatic device design — and the transmon is our demonstration vehicle because its unmet need is the clearest documented in the literature, not because the method is qubit-specific.

A design-gradient claim from a new solver is only as credible as the solver's forward answers, so the paper carries its correctness credential up front: on a real transmon-plus-readout-resonator geometry meshed once and given as the same input file to both codes, geode-fem reproduces Palace — the MFEM-based (Anderson et al., 2021; Andrej et al., 2024) reference solver for superconducting-qubit design — to within 0.033% across the six eigenmodes Palace returns above 4.5 GHz (worst case 0.032%), and to 5×10^{-5} % on the transmon qubit mode at 3.45 GHz, with analytic scaling tripwires guarding against coincidental agreement (Section 5). Palace refines the mesh around the metal sheets internally, so the two discrete problems are not identical (Section 4.1). This parity is *a credential, not the contribution* — matching Palace is the price of entry — and it is, incidentally, the first independent third-party validation of the Palace/DeviceLayout.jl transmon stack, which has no publication of its own.

Our contributions are:

- **A validated 2×2 discrete-adjoint sensitivity matrix for FEM electromagnetics.** Material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$, exact $\partial K/\partial X$ via forward-mode duals through the element kernels) derivatives, on both the scalar electrostatic and the complex driven $H(\mathrm{curl})$ Nédélec operators — each one forward plus one adjoint solve sharing a single factorization, each validated against full-pipeline central finite differences with mutation tests that reject sign, conjugation, and dropped-term errors (Section 8).

- **A differentiable capacitance→ E_C chain with a variational-stationarity simplification.** Because $C = \phi^{\mathsf{T}} K \phi$ is stationary in the solved potential, the implicit term vanishes and the shape derivative is a pure explicit-geometry (Hellmann–Feynman-like) term; validated against the analytic parallel-plate law and central finite differences (Section 8.2).

- **A real-device optimization with a certified two-sided error budget.** An end-to-end Newton loop hits a target E_C on an analytic fixture and is confirmed by an independent fresh solve at 1.4×10^{-15} (the parametrization is affine, so one-step convergence proves the loop, not a hard optimization). On the real device, a three-parameter optimization on an adaptively refined 448k-tetrahedron mesh with quadratic elements reaches the 93.954 fF anchor in four accepted steps, with gradients finite-difference-validated at 4.24×10^{-7} or better. That gradient is a hand-derived equilibrium-energy shape derivative, solved by conjugate gradients with an algebraic-multigrid preconditioner in native f64. It uses the stationarity identity of the previous item, and it runs outside the Burn tape and without the adjoint layer's factorization. Seven held-out refinement rounds give a conforming upper bound, and an equilibrated-flux construction on the same meshes gives a certified lower bound: the converged capacitance of the design lies in $[92.75, 93.00]$ fF, and that of a resolution-corrected design in $[93.89, 94.14]$ fF (Section 9.4).

- **A decomposed design anchor.** The target itself is corrected in two measured steps — the qubit–resonator coupling and a series loop inductance of the junction's current path — from the dressed 93.954 fF to a bare-qubit 93.853 fF and a loop-corrected 93.143 fF. Both steps are extrapolations, not bounds, and rest on a stated transfer assumption. The certified brackets are narrower than the anchor's own uncertainty (Section 9.5).

- **A measured mesh-convergence correction.** In the scalar-permittivity model, the committed 133k-tetrahedron mesh overstates C_Σ by at least 54.69 fF, and the converged unmodified device lies below 83.02 fF under uniform refinement and in $[77.40, 77.61]$ fF on adapted meshes. With the sapphire tensor that the lumped-element chain uses, the overstatement is at least 47.18 fF and the converged value lies below 89.50 fF. Both bounds are under the anchor, where earlier versions placed the device above it. Their "model difference" diagnosis is withdrawn, and the coarse-mesh optimizations that lowered C_Σ toward an anchor are re-read as a cautionary precursor (Sections 7 and 9.3).

- **The correctness credential.** Same-input-mesh, same-physics, same-junction-model cross-validation against Palace at $\leq 0.033\%$ across the six modes above 4.5 GHz and $5 \times 10^{-5}\,\%$ on the qubit mode, with oracle-first methodology and honest-physics reporting — including the correction of two claims made in earlier versions, that the qubit mode was absent and that the 3.45 GHz mode was spurious (Sections 5 and 6) — doubling as the first independent validation of the Palace/DeviceLayout.jl stack.

- **Mode-verified performance context.** Same-box CPU measurements disclose mode selection, field-output and error-estimator costs; they replace the earlier unsupported one-core-versus-eight-rank efficiency claim. CPU-f64 AMS reduces driven-solve time relative to Jacobi, but its iteration and setup costs still grow with refinement (Section 11). Eigenmode/EPR derivatives remain roadmap (Section 10).

Section 2 positions the work; Section 3 presents the substrate and where its tape breaks; Sections 4–6 establish the benchmark problem and the correctness credential; Sections 7–9 are the contribution — the forward LOM chain with its mesh-convergence study, the adjoint layer, and the optimization tiers; Section 10 is the eigenmode roadmap; Sections 11–13 report performance context, reproducibility, and discussion.

2 Related Work

Superconducting-qubit design optimization. The design loop this paper attacks is documented across the qubit-design software stack. QDesignOptimizer (Eriksson et al., 2025) is the state of practice: physics-guided multi-parameter optimization that wraps non-differentiable EM simulation (HFSS with pyEPR/Qiskit-Metal-class post-processing) and steers it with analytic models, because no gradient is available from the solver. The SQcircuit line (Rajabzadeh et al., 2023, 2024) differentiates at the lumped-circuit level — including eigenvalue/eigenvector gradients of the circuit Hamiltonian via PyTorch — and optimizes in circuit-parameter space; the geometry-to-parameters map remains simulation-bound. The SQuADDS database (Shanto et al., 2024) attacks the same loop by lookup: pre-simulated, experimentally-validated design points. Qiskit Metal's quasi-lumped methods (Minev et al., 2021b) and the EPR/black-box quantization physics (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) frame the parameter-extraction layer this paper's LOM chain implements. None of these obtains gradients from the field solve itself; that is the gap Section 8 closes.

Differentiable electromagnetics — owned by photonics. Adjoint/autodiff EM is a mature inverse-design discipline in nanophotonics (Molesky et al., 2018), expressed through structured grids and integral methods: autograd FDFD/FDTD in ceviche (Hughes et al., 2019), foundry-constrained JAX inverse design (Schubert et al., 2022), the Meep adjoint pipeline (Hammond et al., 2022), the JAX-native FDTDX framework (Mahlau et al., 2026), GPU-accelerated autodiff integral-method

scattering in TorchGDM (Ponomareva et al., 2025), and differentiable RCWA (Kim et al., 2024). These codes accept rectilinear grids or Green's-function discretizations; the superconducting-qubit regime — body-fitted tetrahedra, anisotropic sapphire, $H(\mathrm{curl})$ conformity, frequency domain — is exactly where that literature stops. Differentiable FEM has been proven outside EM: JAX-FEM (Xue et al., 2023) (GPU-accelerated, autodiff-native continuum mechanics) and PyTorch-FEA (Liang et al., 2023) in biomechanics; concurrent work on general differentiable Galerkin assembly (TensorGalerkin (Wen et al., 2026), with the torch-sla adjoint sparse-solver library (Chi and Wen, 2026)) independently confirms the substrate-level thesis while reporting nodal magnetostatics rather than full-wave $H(\mathrm{curl})$ EM. To our knowledge, no prior work delivers field-level design gradients from a frequency-domain 3D FEM electromagnetics solve of the class RF/superconducting design uses.

FEM kernels by code generation, and tensor compilers. Generating FEM kernels from high-level descriptions is a two-decade tradition — FFC (Kirby and Logg, 2006), UFL (Alnæs et al., 2014), Firedrake (Rathgeber et al., 2017), TSFC (Homolya et al., 2018) — whose modern GPU expression is libCEED (Brown et al., 2021) and MFEM's partial-assembly path (Andrej et al., 2024). Palace itself runs on that domain-specific stack. geode-fem's substrate bet is categorically different: outsource kernels, scheduling, and backend portability to a *general-purpose ML tensor stack* (the Halide/TVM compiler lineage (Chen et al., 2018; Ragan-Kelley et al., 2013), reaching scientific computing via ML frameworks), and phrase assembly and matrix-free application as batched tensor algebra. For this paper the decisive property of that choice is not portability but *differentiability*: an autodiff-capable substrate makes the assembly derivative available by construction, which is what the adjoint layer of Section 8 composes with. Adjacent genera — simulation DSLs (Holl et al., 2020; Hu et al., 2019), neural-ansatz eigensolvers (Jiang et al., 2024), hand-written CUDA FEM (Meng et al., 2014) — trade away either the general-purpose stack or the discretization itself.

Validation and cross-solver benchmarking. Code-to-code comparison on precisely specified problems is an established genre in computational electromagnetics (the TEAM workshop tradition since 1986[1]; nano-optics solver comparisons (Hoffmann et al., 2009)); in V&V vocabulary (Oberkampf and Roy, 2010; Roache, 1997) our credential is verification-side, code-to-code evidence. On the qubit-EM axis the closest works are SQDMetal (Sommers et al., 2025) (Palace cross-checked against COMSOL/HFSS at the 0.02–0.13% level, independently constructed models) and the Palace-centered workflow of Ye et al. (2025) (validated against cryogenic measurement at ∼0.3%). Both wrap and trust Palace; neither is a same-mesh comparison against an independently *implemented* solver. Our protocol removes mesh generation and model entry as confounders — identical MSH fixture, identical lumped-shunt junction, first-order Nédélec on both sides — so residuals reflect formulation, solver and Palace's internal refinement of the metal sheets (Section 4.1), not independently built models. Mature open EM tools normally carry a methods paper (GetDP (Dular et al., 1998), FEniCS/DOLFIN (Logg and Wells, 2010), Gmsh (Geuzaine and Remacle, 2009), scikit-rf (Arsenovic et al., 2022), MFEM (Anderson et al., 2021)); Palace and DeviceLayout.jl are unpublished outliers (AWS Center for Quantum Computing, 2025; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson, 2025), which is the gap the credential narrows from the outside. geode-fem's numerics: first-order Nédélec elements (Nédélec, 1980), shift-invert Lanczos (Lehoucq et al., 1998) over a real symmetric pencil factorized with faer (Kazdadi, 2026); Palace: SLEPc-class Krylov (Hernandez et al., 2005) with divergence-free projection. The timed Palace build resolves its default linear solver to SuperLU

[1]The TEAM founding documents are workshop reports without resolvable DOIs; we describe the tradition in prose.

under GMRES; explicit AMS runs are reported separately (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009); the gauge-fixing literature behind Section 6 is the tree–cotree tradition (Albanese and Rubinacci, 1988; Manges and Cendes, 1995).

3 GEODE-FEM: A Tensor-Native Substrate, and Where the Tape Breaks

geode-fem is written in Rust on the Burn tensor framework (Tracel AI, 2024), whose cubecl JIT compiles tensor kernels for CPU, CUDA, and WebGPU backends from one source. Every throughput-critical FEM stage is phrased as the batched dense algebra the stack already optimizes.

Batched element-local assembly. For first-order Nédélec tetrahedra the element curl–curl and mass contributions are 6×6 matrices depending on per-element geometry factors and material tensors (including the rotated anisotropic sapphire permittivity of Section 4.1); the scalar electrostatic operator is the analogous P1 kernel. geode-fem computes them for all elements at once as batched $[n_{\text{elem}}, 6, 6]$ tensor operations — one kernel launch per assembly stage rather than an element loop — then either scatters into a global sparse matrix (the assembled f64 CPU path) or retains the batch for matrix-free application.

Matrix-free application and on-device Krylov. The matrix-free operator apply is a three-stage tensor pipeline — gather global DOFs to element-local vectors, batched matmul against the stored element operators, scatter-add back — and the driven frequency-sweep solve runs a conjugate orthogonal conjugate gradient (COCG) iteration entirely on the device with a constant number of scalar host–device synchronizations per iteration. The backend (ndarray on CPU, CUDA on GPU) is a type parameter, not a code path. Section 11 reports honestly how far below the GPU-amortization threshold the current benchmark sizes sit.

Honest constraints. Two boundaries are constraints, not footnotes. The CUDA backend (burn-cuda 0.21) is f32-only today — the underlying cubecl runtime currently disables f64[2] — so every GPU result is mixed-precision-qualified. And the eigensolve that produces the credential comparison is factorization-bound: its inner solves are faer sparse-LU factorizations (Kazdadi, 2026), a classical serial CPU component outside the tensor-compiler story.

Where the tape breaks — and why an adjoint layer, not autodiff-through-the-solver. The substrate choice has a consequence the rest of this paper builds on: Burn's automatic differentiation reaches everything *up to* the linear solve — material and node-coordinate dependence of the assembled operators is differentiable by construction — but the sparse factorization is an external, opaque kernel that breaks the tape. Differentiating *through* a factorization by unrolling is neither practical nor necessary: the classical discrete-adjoint identity turns the missing derivative into one additional linear solve against the *same* factorization (Section 8). The division of labor is deliberate — autodiff where the substrate provides it (element kernels, via forward-mode duals), adjoint algebra where it does not (the solve) — and it is what makes the adjoint-layer gradients of Section 8 and the coarse-mesh tiers of Section 9 *solver-derived* in the sense the design loop of Section 1 lacks: the same assembly and the same factorization that produce the observable produce its gradient. The refined real-device optimization of Section 9.4 is solver-derived in a narrower sense. Its gradient is evaluated from the same solved field that gives the capacitance, but with hand-derived quadratic-element integrals and an iterative solve in native f64; it uses neither the Burn tape nor a factorization.

[2]Upstream constraint of the burn/cubecl stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

4 Benchmark Problem: Geometry, Mesh, and Junction Model

4.1 Geometry and mesh provenance

The device is the `SingleTransmon` example shipped with DeviceLayout.jl v1.15.0 (AWS Center for Quantum Computing, 2025): a transmon with readout resonator on a sapphire substrate, enclosed in a vacuum box. The sapphire is anisotropic, with permittivity tensor $\varepsilon = R\,\mathrm{diag}(9.3,\, 9.3,\, 11.5)\,R^{\mathsf{T}}$ rotated approximately 36.87° in-plane. The DeviceLayout.jl workflow emits seven named physical groups (metal, substrate, vacuum, junction lumped-element surface, two readout ports, exterior boundary) and drives Gmsh (Geuzaine and Remacle, 2009) to produce a tetrahedral mesh, exported as an MSH 4.1 fixture and pinned by SHA-256 (Section 12).

The mesh has 22,684 nodes and 133,314 tetrahedra, yielding 156,863 Nédélec edge unknowns, of which 133,108 interior degrees of freedom remain after eliminating perfect-electric-conductor (PEC) boundary edges. PEC is applied on all metal surfaces and the exterior box; the two readout ports are left open (lossless), so no resistive element enters and the eigenproblem pencil stays real symmetric. These counts are geode-fem's. Palace reads the same file but does not solve the same discrete problem: its run logs record 4,913 elements added by local bisection around under-resolved interior boundaries and 6,014 duplicated vertices where it cracks the metal sheets, and it assembles 183,891 first-order Nédélec unknowns against geode-fem's 156,863 (about 17% more). "Same mesh" in this paper therefore means the same *input* mesh. Earlier versions said the two solvers consumed an identical mesh and solved an identical discrete problem; that was wrong, and the timing comparisons of Section 11.1 carry the difference as a fairness caveat. This mesh is also coarse for the electrostatic quantities: one uniform refinement lowers C_Σ by 26.3% (Section 7). Figure 1 shows the geometry with physical groups color-coded. Every measurement in this paper starts from this one committed mesh. The eigenmode credential and the coarse-mesh electrostatic chain run on it directly; the scale study of Section 11 uses a uniformly refined variant; and the convergence study and refined optimization of Sections 7 and 9.4 use uniform and adaptive refinements of it that leave its polygonal surfaces unchanged.

4.2 The junction model: a lumped reactive shunt

The Josephson junction is modeled identically in both solvers as a lumped reactive shunt — a parallel $L \parallel C$ element with $L = 14.860\,\mathrm{nH}$ and $C = 5.5\,\mathrm{fF}$ — attached to the four-triangle `lumped_element` surface group Γ. Following the uniform lumped-port construction, with port length ℓ and width w, the shunt contributes

$$K_{\mathrm{port}} = \frac{\ell}{w\,L}\,S_\Gamma \qquad \text{and} \qquad M_{\mathrm{port}} = \frac{C\,\ell}{w}\,S_\Gamma, \tag{1}$$

where S_Γ is the port-surface matrix assembled on Γ. The eigenproblem is then the real symmetric generalized pencil

$$(K + K_{\mathrm{port}})\,x \;=\; \omega^2\,(M + M_{\mathrm{port}})\,x. \tag{2}$$

In Palace, the same physics is entered as a reactive `LumpedPort` ($L \parallel C$, no resistance) on the same surface group; the readout ports are configured open, so Palace's pencil is likewise real. Palace's source confirms the correspondence: its `LumpedPort` adds the same scalar surface-mass terms with the same per-square scalings (`docs/research/transmon-lumped-port-formulation.md`, §1). This one-to-one correspondence is what makes the comparison same-physics as well as same-input-mesh. It also means that any property of the distributed surface term in Eq. 1 is shared by both solvers (Section 6.3).

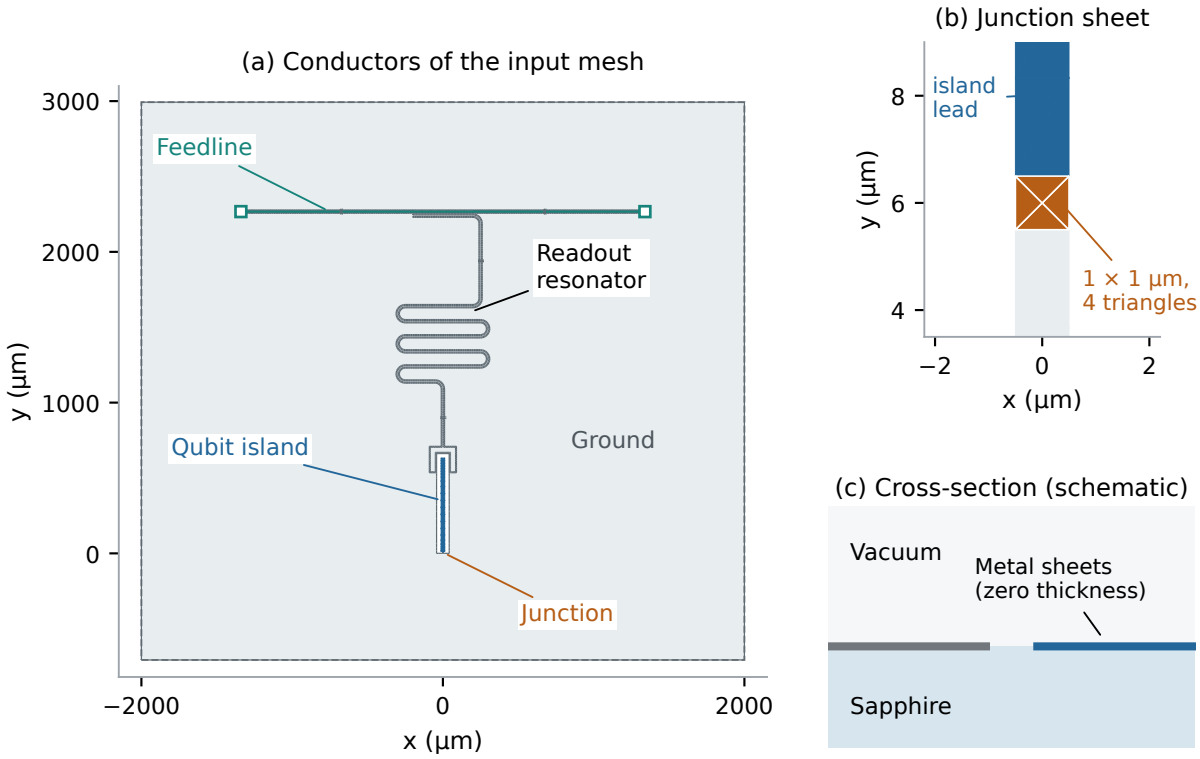


Figure 1: Benchmark geometry, drawn from the committed SHA-256-pinned MSH 4.1 input mesh of the DeviceLayout.jl v1.15.0 `SingleTransmon` device. (a) Top view of the three node-disjoint metal conductors of Section 7: the feedline (teal), the qubit island (blue), and the ground plane together with the meander readout resonator (gray), whose etched gaps are drawn as dark outlines. Squares mark the two readout ports; the dashed line is the exterior-boundary outline. (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its actual four-triangle triangulation. (c) Material cross-section, schematic and not to scale; the `substrate` and `vacuum` volume groups fill the enclosing box. Both solvers read this file (22,684 nodes; 133,314 tetrahedra); Palace then refines it internally around the metal sheets, so the two do not solve the same discrete problem (Section 4.1).

5 The Correctness Credential: Same-Input-Mesh Cross-Validation Against Palace

A solver claiming design gradients must first prove its forward answers; this section is that proof. Palace's own agreement with commercial tools is established (Sommers et al., 2025), so reproducing Palace is the bar a new implementation must clear before anything it computes *about* its solutions (such as their derivatives) is worth discussing.

Protocol. The benchmark is oracle-first. The mesh is generated once, committed, pinned by SHA-256, and read by both solvers at matched (first) order with the identical junction model; residual disagreement is therefore attributable to formulation, implementation or Palace's internal refinement (Section 4.1), not to independent meshing. Two analytic tripwires guard against coincidental or self-consistent-but-wrong agreement. Both test the port term's L and C scalings. The first is the

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes / 133,314 tets; geode-fem solves 156,863 Nédélec DOFs, 133,108 interior after PEC; Palace refines internally and solves 183,891), first order in both solvers, junction as a lumped reactive shunt ($L = 14.860\,\text{nH}$, $C = 5.5\,\text{fF}$). These are the six modes Palace returns when it searches above 4.5 GHz; the qubit mode below that target is in Table 2. geode-fem at commit `3174015`; Palace at commit `fba6a5b`. All values are quoted verbatim from the committed `benchmarks/transmon_eigen/results.toml`: geode-fem frequencies at the committed three decimals, Palace at full precision, and the committed per-mode `rel_err_pct`, computed against the rounded geode-fem values (precision disclosure in Section 12).

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
`resonator`	readout resonator	5.153	5.151335830348	0.032
`mode_2`	box/cavity	15.465	15.46052107794	0.029
`port_sheet_artifact`	port-sheet LC*	17.490	17.49010903536	0.001
`mode_4`	box/cavity	18.693	18.69165792915	0.007
`mode_5`	box/cavity	20.703	20.69755679425	0.026
`mode_6`	box/cavity	26.088	26.08089940472	0.027

*The artifact key was `junction_lc` before the re-baselining of issue #1045 (PR #1071); the value is unchanged. Palace's port EPR for this mode is 2.49×10^{-8}, and Section 6.3 identifies it as an artifact of the distributed port term, which the opt-in rank-1 shunt removes.

resonance of the port sheet's own inductance against its own capacitance, $f_{LC} = 1/(2\pi\sqrt{LC}) = 17.60\,\text{GHz}$, which anchors the 17.49 GHz mode. The second is a $\sqrt{L}$-scaling test: doubling L must move that mode by $1/\sqrt{2}$ — measured: 17.4901 → 12.37 GHz, committed ratio 0.7071 (computed from the unrounded values; Figure 3). Earlier versions called the 17.49 GHz mode "the junction mode" and treated these two checks as pinning the junction physics. Section 6.3 shows that this mode carries no port voltage, so the checks pin the port *operator*, not the qubit; the qubit mode has its own scaling check in Section 6.1. The suite also contains inverse tests that *must fail* (perturbed configurations required to disagree with the oracle — a pipeline that still passes under them is not measuring anything), and the Palace oracle is deterministic: a rerun from the committed configuration reproduced its committed `eig.csv` bit-for-bit.

Mode identification. Modes are identified, not eyeballed. Within geode-fem, the stiffness-participation ratio

$$p = \frac{x^{\mathsf{T}} K_{\text{port}}\, x}{x^{\mathsf{T}} (K + K_{\text{port}})\, x} \tag{3}$$

measures the share of a mode's stiffness energy stored in the port surface term, whatever its spatial pattern. It is $p = 1.000$ for the 17.49 GHz mode, $p = 0.9942$ for the 3.45 GHz mode, and $p = 0.000$ (at the artifact's three-decimal precision) for the resonator and box modes. Palace's port EPR is built from the port *voltage* functional alone, so it counts only the uniform-voltage part of that energy. The two diagnostics agree where the port field is a uniform voltage and disagree where it is not: on the qubit mode Palace reports 0.99424 against geode-fem's 0.9942, while on the 17.49 GHz mode Palace reports 2.49×10^{-8} against geode-fem's 1.000 (Section 6). In the original Palace run, which searched only above 4.5 GHz, all six port-EPR values are $\leq 5 \times 10^{-4}$; earlier versions read that as two diagnostics ranking the modes differently, when the cause was the qubit mode's absence from Palace's search window. The six-mode comparison of Table 1 is matched by frequency.

Result. Table 1 and Figure 2: all six modes agree to $\leq 0.033\%$ (worst case 0.032%). The 17.49 GHz port-sheet mode agrees most closely, at 0.001%, and sits 0.6% below $f_{LC} = 17.60\,\text{GHz}$ in *both*

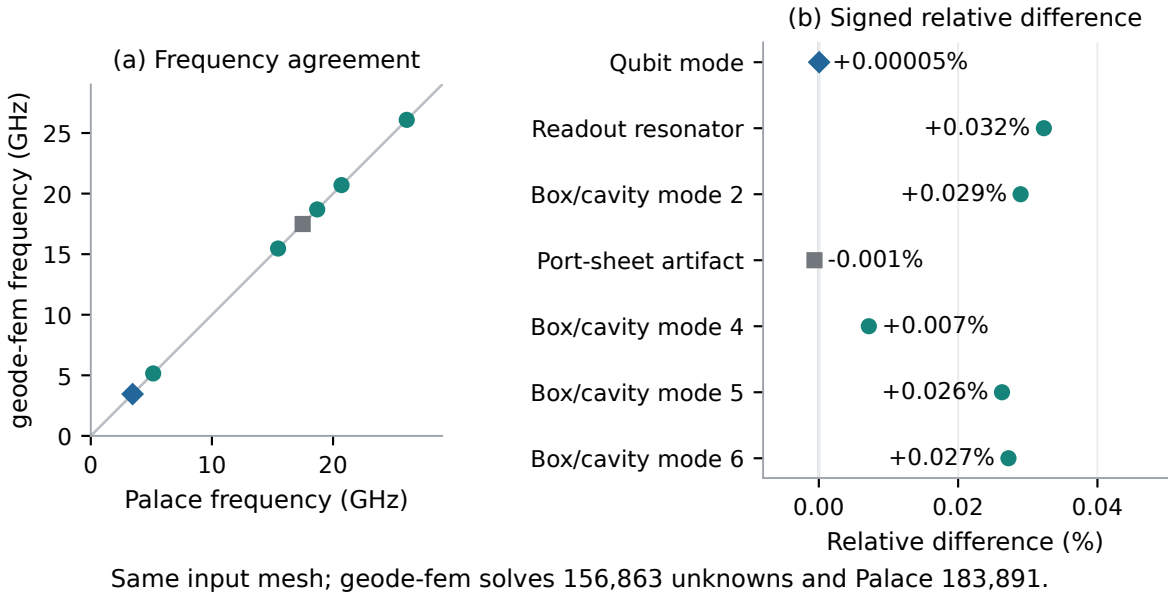
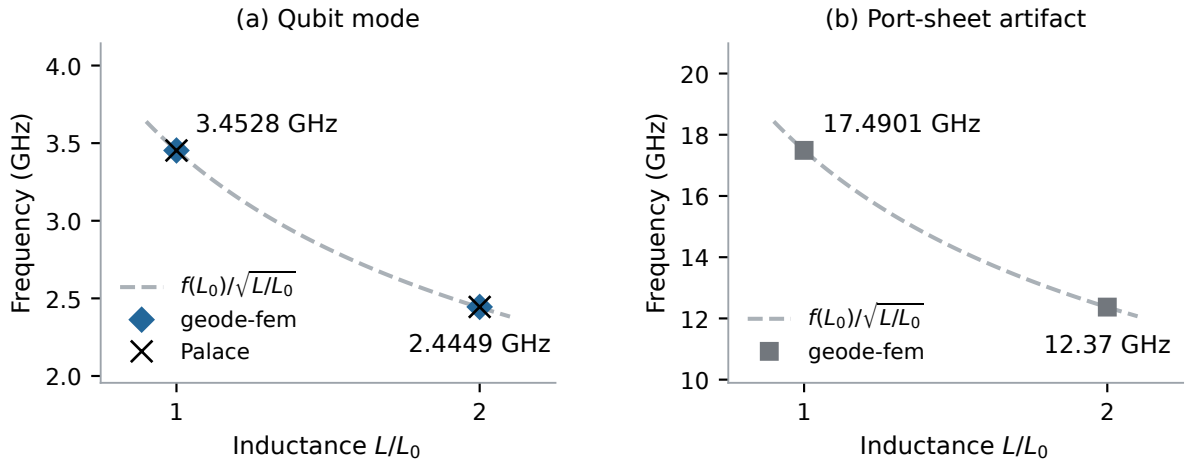


Figure 2: Mode-frequency agreement from the identical input mesh: the six modes Palace returns above its 4.5 GHz search target (Table 1) and the qubit mode from the `Target` = 2 GHz re-run (Table 2). (a) geode-fem against Palace frequency, with the identity line. (b) Signed relative difference $100\,(f_{\text{geode}}/f_{\text{Palace}} - 1)$, computed from the committed frequencies: geode-fem at its committed three decimals for the six modes, which reproduces the per-mode values of Table 1, and at its committed six decimals for the qubit. The 17.49 GHz port-sheet artifact is marked separately; agreement on it tests the shared port operator, not the qubit physics (Section 6.3). Data from `benchmarks/transmon_eigen/results.toml`.

solvers. Agreement that tight on a mode confined to the four port triangles is what an identical port operator would produce, and it is weaker evidence about the bulk discretizations than the cavity modes' ∼0.03%. The qubit mode, which Palace returns only when asked to search below it, agrees to 5×10^{-5} % (Section 6.1, Table 2). The agreement is between genuinely independent code paths — a batched-tensor assembly feeding a direct-factorization Lanczos in Rust, and a projected, preconditioned Krylov method in C++ — on discrete problems that differ by Palace's internal refinement (183,891 against 156,863 unknowns). A common implementation error is therefore an unlikely explanation for the residual. A shared *modeling* property is not excluded by cross-comparison, and Section 6.3 reports one. All of this is same-input-mesh agreement at first order; none of these frequencies is a mesh-converged value.

6 Honest Physics: The Qubit Mode, Two Withdrawn Claims, and a Port-Sheet Artifact

Versions 1–6 of this paper made two claims in this section, and both were wrong. The first was that no qubit mode exists in this junction model by construction, and that Palace confirmed its absence. The second was that geode-fem's 3.4528 GHz mode is a spurious port artifact with no Palace counterpart. We withdraw both. The committed evidence identifies the 3.45 GHz mode as the transmon qubit mode, which Palace also returns, and it points to a different mode, at 17.49 GHz, as the artifact of the port model. This section separates what is proven by committed artifacts from what is derived, what a scratch probe measured, and what is only planned.



$L_0 = 14.860$ nH. Qubit mode: 0.70810 (Palace, from its committed frequencies), 0.7081 (geode-fem, committed). Port-sheet artifact: 0.7071 (geode-fem, committed). $1/\sqrt{2} = 0.70711$.

Figure 3: Inductance-doubling checks with $L_0 = 14.860$ nH for (a) the qubit mode and (b) the 17.49 GHz port-sheet artifact. Dashed curves show $f(L_0)/\sqrt{L/L_0}$, anchored at each mode's own baseline. Doubling L moves the port-sheet mode from 17.4901 to 12.37 GHz (committed ratio $0.7071 = 1/\sqrt{2}$), and moves the qubit mode from 3.4528 to 2.4449 GHz in geode-fem (ratio 0.7081) and from 3.452749 to 2.444899 GHz in Palace (ratio 0.70810); at this scale the Palace markers cover geode-fem's. The qubit mode's ratio sits slightly above $1/\sqrt{2}$ because the full-wave mode lies 0.28% below the lumped estimate at L and 0.14% below it at $2L$, an offset that a series loop inductance mostly accounts for (Section 6.1). Both modes respond to L, so this check alone does not identify the physical qubit mode. Palace's 0.70810 is computed from its two committed frequencies; the geode-fem ratios are committed fields. Values from the committed `benchmarks/transmon_eigen/results.toml` (`[tripwires.port_sheet_artifact_l_doubling]`, `[tripwires.qubit_l_doubling]` and `[oracles.palace_target2]`).

6.1 The 3.45 GHz mode is the transmon qubit mode

Why the earlier Palace run could not return it. The original Palace run asked for six modes with `Target` = 4.5 GHz and returned nothing below the 5.15 GHz resonator. Earlier versions read that as Palace hunting near 4.5 GHz and finding no qubit mode. That misreads `Target`. Palace's eigensolver accepts only eigenvalues at or above the target: its SLEPc configuration restricts the search region to that side, and its configuration schema defines `Target` as the frequency "above which to search" (`docs/research/transmon-lumped-port-formulation.md`, §2, read from the Palace source at the benchmarked changeset). The run's silence below 4.5 GHz was therefore not evidence of absence.

Proven: Palace returns the mode when asked. We re-ran Palace at the same changeset on the same input mesh, with the same materials, `LumpedPort`, order, eigensolver tolerances and linear solver, changing only `Target` (4.5 → 2.0 GHz) and the number of modes (6 → 8); a second cell also doubles L. A control cell with the original request reproduces the committed spectrum to a relative 6.5×10^{-13}, so the local build does not move the eigenvalues. Table 2 gives the result (`benchmarks/transmon_eigen/results.toml`, `[oracles.palace_target2]`; raw outputs under `reference/fixtures/transmon_palace/results_p1_target2/`).

Table 2: The qubit mode in both solvers, from the identical input mesh at first order. Palace values are from the `Target` = 2 GHz re-run; geode-fem values are the committed ones that earlier versions reported as a spurious mode. The lumped estimate is $1/(2\pi\sqrt{L(C_\Sigma + C_J)})$ with the electrostatic $C_\Sigma = 136.684731$ fF of Section 7 and $C_J = 5.5$ fF. Palace's column is its port EPR; geode-fem's is the stiffness participation of Eq. 3. This is a same-input-mesh comparison on a coarse mesh, not a converged qubit frequency.

Quantity	Palace	geode-fem	Lumped estimate
Qubit mode, $L = 14.860$ nH (GHz)	3.452749	3.452751	3.46246
Port EPR / participation p	0.99424	0.9942	—
Qubit mode, $2L$ (GHz)	2.444899	2.4449	2.44833
L-doubling ratio ($1/\sqrt{2} = 0.70711$)	0.70810	0.7081	0.70711
17.49 GHz mode: port EPR / p	2.49×10^{-8}	1.000	—
17.49 GHz mode at $2L$ (GHz)	12.3674	12.37	—

Palace finds the mode at 3.452749 GHz with port EPR 0.99424; geode-fem's committed value is 3.452751 GHz with $p = 0.9942$. The two frequencies agree to 5×10^{-5} %. Under L doubling the mode moves to 2.444899 GHz in Palace, a ratio of 0.70810, against geode-fem's committed 0.7081. Palace's value lies 0.280% below the lumped estimate at L and 0.140% below it at $2L$. That estimate takes C_Σ from the electrostatic branch on the same coarse mesh, so it is a consistency check between this paper's eigenmode and LOM branches, not an independent anchor. Every other mode in the re-run matches the committed Palace spectrum, and two further modes converge above it (33.384 and 33.896 GHz).

Most of the 0.28% offset is a series loop inductance. Earlier versions left the gap between Palace's qubit mode and the lumped estimate unexplained. A later measurement on the same fixture (`benchmarks/transmon_loop_inductance/results.toml`, issue #1079) finds an inductance L_s in series with the junction, outside the junction's own $L \parallel C$. Two tests say it is physical and say where it sits. Moving the stretch of the junction wire that carries the port sheet, so that the sheet's length changes twelvefold, moves L_s by at most 1.17% at first order and 1.27% at second order, and geode-fem and Palace agree on it to better than 2×10^{-4}; a port-model artifact would follow the sheet. Changing the junction capacitance from 0 to 20 fF leaves the series estimator of L_s constant to 3.3×10^{-4}, while the parallel reading changes by 25.6%. On this mesh $L_s = 0.0852$ nH, and it accounts for 94% of the −0.280% offset; the qubit–resonator hybridization accounts for 5%, and −0.0036% (1%) is left for solver and permittivity-model differences. At $2L$ the loop share halves, as a series inductance predicts, and the residual stays at −0.0032%. This decomposition is on the same coarse mesh as the rest of this section, and L_s is not mesh-converged (Section 9.5). The remaining drift of about 1% with the sheet's position is an observation; it does not bound the port model's share of L_s.

Earlier versions explained the supposed absence by saying that the lumped-shunt model gives the junction only its own 5.5 fF, so that its resonance lands at 17.5 GHz, and that the pad capacitance of a physical transmon (Koch et al., 2007) was missing. That explanation was wrong. The full-wave eigenproblem contains the island's capacitance through the field itself, which is why a mode exists in which the junction inductance resonates against the whole island.

What this does not show. This is not the converged physical qubit frequency. Both solvers start from the same coarse 133k-tetrahedron mesh at first order, and they do not solve the same discrete problem: Palace refines around the metal sheets and solves 183,891 unknowns against

geode-fem's 156,863 (Section 4.1). The mesh is also far from converged for the capacitance that sets this frequency: one uniform refinement lowers the scalar-ε C_Σ by 26.3% (Section 7). The lumped estimate goes as $1/\sqrt{C_\Sigma + C_J}$, so we infer that the qubit frequency is not converged on this mesh either; that inference has not been tested by a refined eigensolve. The DeviceLayout.jl blog workflow (Peairs and Carson, 2025) reports a spectrum from 4.14 to 5.591 GHz that includes a ∼4 GHz qubit mode. We do not claim to reproduce it, and we do not attribute the difference from our 3.45 GHz to any single cause.

The committed artifacts are re-baselined. Since issue #1045 (PR #1071), `benchmarks/transmon_eigen/results.toml` stores this mode as `[modes.qubit]`, with `kind = "qubit"` and no exclusion flag; it was `[spurious_mode]` with `excluded = true`. The 17.49 GHz mode is `[modes.port_sheet_artifact]`, and the CPU timing artifacts class the 3.45 GHz mode as `qubit` where they had `spurious_port`. No measured value changed. The artifacts' header comments carry the full map from old keys to new ones.

6.2 The gauge and projection arc, re-read

Earlier versions told the following arc as the diagnosis of a spurious mode. Its measurements stand, every step committed (Section 12); what they mean changes once the two modes are correctly identified. geode-fem's headline Lanczos operates on the raw pencil of Eq. 2 without a gauge projection. (1) Classical tree–cotree elimination (Albanese and Rubinacci, 1988; Manges and Cendes, 1995) is not spectrum-preserving for the generalized eigenproblem — the resonator shifts 1.64% (5.1528 → 5.2372 GHz), outside our bar, pinned as a regression test. (2) The spectrum-preserving bulk divergence-free projector (structurally Palace's confinement) preserves the cavity spectrum (0.029%) and collapses the 13,747-mode gradient cluster. It removes the 17.49 GHz mode, which is almost entirely a gradient field (measured divergence ratio 50.2; projected norm 1.06×10^{-4}), and it keeps the 3.45 GHz mode, which is solenoidal with respect to the interior gradients (divergence ratio $\approx 6 \times 10^{-15}$). Earlier versions described this as the projector deflating the physical junction mode while the spurious mode survived. With the identities corrected it reads the other way: the projector kept the qubit mode and removed the port-sheet mode. (3) A port-aware rank-1 re-admission of one gradient direction restores the 17.49 GHz mode and returns all six of Palace's `Target` = 4.5 GHz modes at $\leq 0.029\%$ worst case. That construction did what it was built to do. Its purpose, though, was to retain a mode that Section 6.3 identifies as an artifact, because the Palace reference then in hand contained that mode and did not contain the qubit mode.

The remaining committed characterization of the 3.45 GHz mode is unchanged and now reads as a description of the qubit mode: under the L-doubling harness it moves 3.4528 → 2.4449 GHz (ratio 0.7081) with 99.4% of its stiffness energy in the K_{port} term. Earlier versions added that Palace's `LumpedPort` represents the port differently and therefore never sees this mode. That was also wrong: Palace's port operator has the same structure (Section 4.2), and Palace sees the mode.

The headline comparison deliberately remains the committed ungauged benchmark. The subsequent port-subspace implementation removes the near-zero survivor and admits the port-patch gradient direction with one shifted factorization. On this fixture, requesting seven modes at 4.5 GHz returns the six modes of Table 1 (worst relative frequency error 0.0289%) plus the 3.452751 GHz qubit mode; requesting six still misses the highest of the six. The committed local comparison records mean process CPU times of 46.32 s for the older eight-mode extraction route and 23.25 s for the new seven-mode route. These are geode-only measurements on a loaded Apple M3 Ultra, not a Palace comparison or a same-box update to the EC2 timings (Section 11.1). With the default port term, geode-fem's benchmark harness still classifies modes by frequency matching against the

Palace reference. With the opt-in rank-1 shunt of Section 6.3, the plain bulk projector returns the qubit mode and the five cavity modes without a reference filter; that route is measured on this fixture only and is not the default. Figure 4 shows the spectrum.

6.3 The 17.49 GHz mode: a zero-voltage resonance of the distributed port term

Proven by committed data. Palace's port EPR for the 17.49 GHz mode is 2.49×10^{-8}, in the original run and in the re-run. After L doubling the mode moves to 12.3674 GHz (ratio 0.70711) and its port EPR is 2.94×10^{-8}. Palace builds that EPR from the port voltage, so in Palace this mode carries essentially no port voltage. In geode-fem the same mode has $p = 1.000$: all of its stiffness energy is in the port surface term. Both statements hold only if that energy sits in a surface pattern whose net voltage is zero.

Measured by a scratch probe. A probe on the committed fixture measured the uniform-voltage share of each mode's port energy in geode-fem: 0.00% for the 17.49 GHz mode and 100.00% for the 3.45 GHz mode. The probe source is committed (`docs/research/transmon-lumped-port-formulation.probe-1003.rs`) but it is scratch evidence: the build does not compile it, and no regression test pins these two numbers.

Derived. The junction patch is four triangles with five nodes, of which only the centre node is not on a conductor. The gradient of that node's hat function is curl-free and has zero port voltage, but the isotropic surface term of Eq. 1 still stores energy in it. A field confined to the patch then has Rayleigh quotient $1/(LC)$, which is $f_{LC} = 17.60$ GHz, and the volume mass of the adjacent tetrahedra lowers it to 17.49 GHz. A two-terminal junction cannot carry this pattern. On this reading the mode is a sheet-internal resonance created by representing a lumped element as a distributed isotropic sheet, and both solvers have it because both use that representation. The reading predicts one such mode per free patch node, so a finer port mesh should show a cluster just below f_{LC}. That falsifier has not been run.

What it means for the credential. The 0.001% agreement on this mode is agreement on a shared artifact of the port model. It still tests that the two implementations assemble the same port operator, and nothing more. The physical content of the credential is the five cavity modes and the qubit mode.

Built, opt-in: a rank-1 port-voltage shunt. The remedy is now implemented as an opt-in junction model (issue #1044, PR #1067): it replaces S_Γ in Eq. 1 by $b\,b^\mathsf{T}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\, dS$ the port-voltage functional, so that only the energy of the port voltage is kept. A committed release test on the committed fixture (`benchmarks/transmon_eigen/results.toml`, `[port_voltage_a2]`) measures three things. First, $\max|G^\mathsf{T} b| = 0$ exactly, so the plain bulk projector is exact under this shunt. Second, the bulk projector at 4.5 GHz with six modes requested returns the qubit mode at 3.452753 GHz and the five cavity modes, with residuals $\leq 1.9 \times 10^{-9}$, in one factorization and with no Palace frequency filter (the only filter drops one exact null pair). Third, an ungauged solve at 17.5 GHz finds no mode between 15.464979 and 18.692705 GHz, where the shipped shunt has its 17.490109 GHz mode. Every mode it keeps agrees with the shipped shunt to four decimals. The default junction model is unchanged, every table in this paper uses it, and a solver using the new shunt no longer reproduces Palace's 17.49 GHz mode; whether it becomes the default is an open decision (issue #1046). That the removed mode is the sheet-internal resonance derived above is an inference, which a synthetic test supports; the finer-port-mesh falsifier has still not been run.

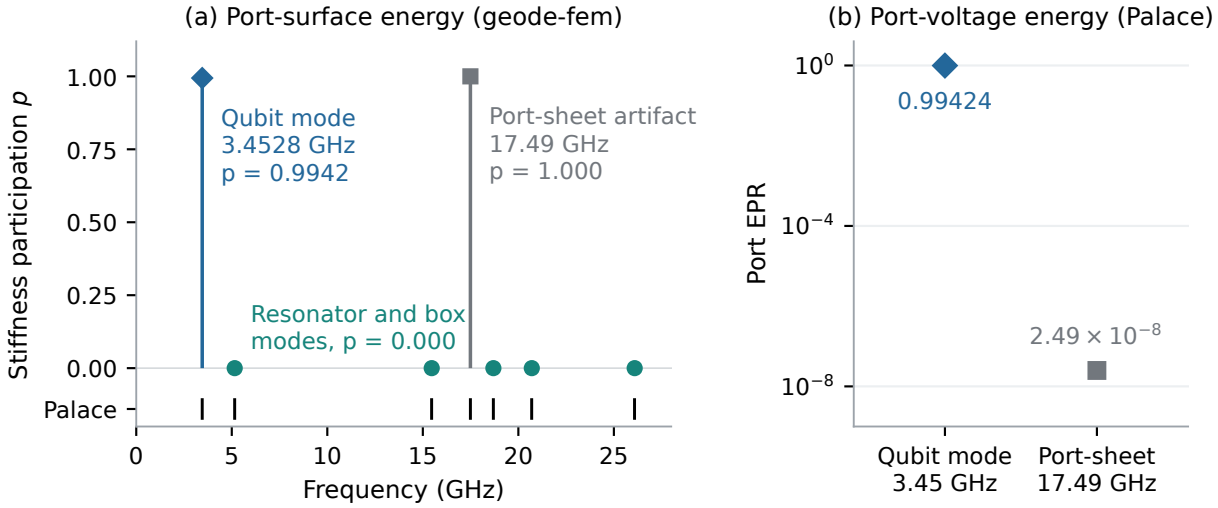


Figure 4: Two different port diagnostics, on separate axes. (a) geode-fem's stiffness participation p (Eq. 3), which counts all port-surface energy, with the mode frequencies of Palace's `Target` = 2 GHz re-run marked below the axis (its two modes above 33 GHz are outside the plotted range). The qubit mode appears at 3.45 GHz in both solvers. It and the 17.49 GHz port-sheet artifact both have participation near one in geode-fem ($p = 0.9942$ and $p = 1.000$); the resonator and box modes have $p = 0.000$. (b) Palace's port EPR, built from the port voltage, on a logarithmic axis: 0.99424 for the qubit mode but 2.49×10^{-8} for the artifact, whose port energy carries no port voltage (Section 6.3). The two quantities coincide only for uniform-voltage port fields (Section 5), so they are not interchangeable measures of participation.

7 The Forward LOM Chain: From Fields to E_C

The lumped-oscillator-model (LOM) branch computes the transmon's Hamiltonian parameters from the *electrostatic* problem on the same committed mesh — and it is the chain the remainder of this paper differentiates. The mesh's metal is split into three node-disjoint conductors (ground+resonator, feedline, island); a tensor-ε electrostatic solve yields the Maxwell capacitance matrix by the energy-reaction form $C_{ij} = \phi_i^{\mathsf{T}} K \phi_j$; the island self-capacitance is reduced for the floating feedline ($C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$); and the charging energy follows as $E_C/h = e^2/(2(C_\Sigma + C_J)h)$, with the junction's $C_J = 5.5\,\mathrm{fF}$ added back (issue #1066; earlier versions omitted it), and the transmon spectrum evaluated charge-basis-exactly after Koch et al. (2007). C_Σ itself excludes C_J throughout this paper.

Measured on the committed fixture (`benchmarks/transmon_quantum/results.toml`): $C_\Sigma = 136.7\,\mathrm{fF}$, hence $E_C = 0.136\,\mathrm{GHz}$, $E_J/E_C = 80.7$ (with E_J from the junction inductance), $\omega_{01} = 3.32\,\mathrm{GHz}$, and anharmonicity $\alpha = -0.151\,\mathrm{GHz}$, with the $\alpha \approx -E_C$ sanity holding. The chain runs end-to-end from committed geometry. These are coarse-mesh numbers and they are not mesh-converged, as the next paragraph shows. The harmonic frequency of the same lumped circuit, $\sqrt{8E_JE_C} = 1/(2\pi\sqrt{L(C_\Sigma + C_J)}) = 3.4625\,\mathrm{GHz}$, is the lumped estimate of Table 2, which the linear eigenproblem resolves at 3.45 GHz (Section 6.1); ω_{01} is the anharmonic $0 \to 1$ transition and lies about E_C below it.

Table 3: Mesh convergence of C_Σ (fF) for the unmodified device under uniform refinement of the committed mesh. Every entry is a conforming energy minimum and therefore an upper bound on the converged value for the committed polygonal geometry. No P2 solve was run at $h/4$, and the tensor-ε model has no P2 path. Source: `benchmarks/transmon_diffopt/c_sigma_convergence.toml`.

Level	Tetrahedra	P1, scalar ε	P2, scalar ε	P1, tensor ε
h	133,314	137.706752	90.044141	136.684731
$h/2$	1,066,512	101.359397	83.015285	102.375301
$h/4$	8,532,096	87.753422	n/a	89.493342

Mesh convergence of C_Σ. A convergence study on the committed geometry (`benchmarks/transmon_diffopt/c_sigma_convergence.toml`) refines the committed mesh uniformly, splitting every tetrahedron into eight, and re-solves C_Σ of the unmodified device, written X^0 below. It uses the scalar-ε model, the one the shape gradient differentiates, with linear (P1) elements at h, $h/2$ and $h/4$ and quadratic (P2) elements at h and $h/2$ (Table 3, Figure 5). The $h/4$ mesh has 8,532,096 tetrahedra. A sparse direct factorization does not fit safely at that size on the 96 GiB host, so $h/4$ and P2 at $h/2$ are solved by conjugate gradients with an algebraic-multigrid preconditioner; where both solvers run, they agree to a relative 10^{-9} or better.

Rigorous upper bounds. The P1 spaces are nested under the same Dirichlet and floating constraints, the P2 space on a mesh contains the P1 space on it, and C_Σ is an energy minimum. Every solved value is therefore an upper bound on the converged C_Σ of this geometry. The smallest, P2 at $h/2$, puts the converged scalar-ε value of X^0 below 83.02 fF. The coarse 137.706752 fF overstates it by at least 54.69 fF, which is at least 65.8% of the converged value.

Certified lower bounds. A complementary-energy construction now supplies the other side (issue #1062, PR #1072; `benchmarks/transmon_diffopt/c_sigma_bracket.toml`). For any piecewise-polynomial displacement field D on the mesh cut along the sheets, Thomson's principle and the Cauchy–Schwarz inequality give $C_\Sigma \geq \varepsilon_0 N^2 / \int D \cdot \varepsilon^{-1} D$, where N is the island charge minus the absolute charge on the floating feedline and the L^1 norms of every normal-flux jump off the sheets and of $\operatorname{div} D$. The bound holds for any D, so solver tolerance and roundoff can only weaken it. The field used is the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three (Prager–Synge, Braess–Schöberl), with an independent Crouzeix–Raviart flux as a cross-check; it is validated against an exact parallel plate, an analytic coaxial bracket and closed-form checks of each certificate term, with mutation tests that reject a grounded-feedline or jump-free certificate. On the uniform levels the bound is weak: P2 at $h/2$ gives only $[74.05, 83.02]$ fF. On the adapted meshes of Section 9.4 it is tight, and the converged scalar-ε C_Σ of X^0 is certified in $[77.40, 77.61]$ fF, a bracket 0.21 fF wide. Certified means rigorous for the discrete polygonal geometry, with lower bounds floored and upper bounds ceiled; the certificate is evaluated in plain f64, not interval arithmetic, with a worst-case summation error the artifact puts far below the rounding.

Estimates, now ruled out. Four extrapolations span 74.14–79.62 fF: Richardson extrapolation of the P1 levels at the observed order and at order one, a fit of $C_\infty + ah + bh^2$ through the three P1 levels, and Richardson extrapolation of the two P2 levels at order one. The observed P1 order is 1.41. We expected order one, because the potential near the edge of a zero-thickness sheet behaves like $r^{1/2}$; that expectation is an inference from theory and not a measurement. The band is not asymptotic and it is not a bound, and each of its four values lies outside the certified bracket. The P2 value on the committed mesh, 90.044141 fF, is not converged either: one refinement lowers it by a further 7.02 fF.

Tensor permittivity. With the sapphire tensor, the P1 values in Table 3 fall in the same way.

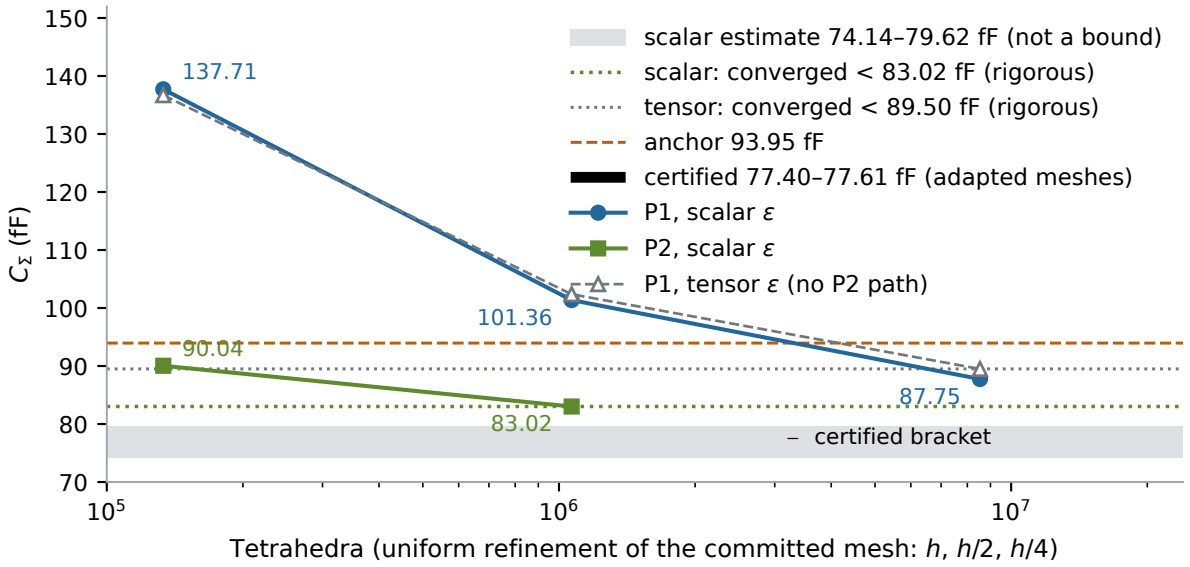


Figure 5: Mesh convergence of C_Σ for the unmodified device X^0 under uniform refinement of the committed mesh (Table 3). Every plotted marker is a conforming upper bound on the converged value of the committed polygonal geometry; the shaded band is an extrapolated estimate and is not a bound. Scalar permittivity: P1 at h, $h/2$ and $h/4$ and P2 at h and $h/2$, with the rigorous upper bound of 83.02 fF and the estimate band of 74.14–79.62 fF. Tensor permittivity, which has no P2 path: P1 only, converged below 89.50 fF. The black bar is the certified two-sided bracket $[77.40, 77.61]$ fF of the scalar value, from the equilibrated-flux lower bound on the adapted meshes of Section 9.4, drawn at the size of the mesh that sets it; it rules out every point of the estimate band outside it. The dashed line is the 93.95 fF anchor. Source: `benchmarks/transmon_diffopt/c_sigma_convergence.toml` and, for the bracket, `c_sigma_bracket.toml`.

The converged tensor C_Σ is below 89.50 fF, and two Richardson values put it at 76.61–81.75 fF, an estimate. Its certified lower bound, from the uniform P1 levels only, is 72.96 fF, so the tensor bracket is $[72.96, 89.50]$ fF, far wider than the scalar one; the tensor model has no P2 or adaptive path. The scalar-minus-tensor difference changes sign under refinement: it is $+0.75\%$, -0.99% and -1.94% of the tensor value at h, $h/2$ and $h/4$. The 0.75% scalar-versus-tensor difference is thus a coarse-mesh number and not a converged model difference.

Scope. "Converged" means converged for the polygonal surfaces of the committed mesh, which refinement does not move. It is not a statement about a regenerated or re-meshed DeviceLayout device.

Consequences for the chain. Every C_Σ, E_C, E_J/E_C, ω_{01} and α quoted at the start of this section is a coarse-mesh number. The coarse tensor C_Σ overstates the converged one by at least 47.18 fF. With C_J included, as the chain now does, the converged E_C is therefore at least 49.6% above the coarse 0.136 GHz (our computation from the committed bound; the artifact's companion figure of 52.7% treats $E_C \propto 1/C_\Sigma$ without C_J). The Koch spectrum has not been re-evaluated at a converged C_Σ.

The design anchor, and why it changed. The anchor for the island capacitance is the published DeviceLayout.jl qubit frequency, 4.14 GHz, the lower mode of the pair $[4.14, 5.591]$ GHz that its SingleTransmon example reports (Peairs and Carson, 2025). Palace computes that number as a

linear eigenmode with the junction as the lumped $L = 14.860\,\mathrm{nH}$ in parallel with $C_J = 5.5\,\mathrm{fF}$, so it is a harmonic frequency, $f = 1/(2\pi\sqrt{L(C_\Sigma + C_J)})$. Inverting it gives $C_\Sigma + C_J = 99.4537\,\mathrm{fF}$, a target $C_\Sigma = 93.9537\,\mathrm{fF}$ with the junction excluded, and $E_C = 0.194766\,\mathrm{GHz}$ (`benchmarks/transmon_quantum/results.toml`, `[quantum.anchors]`; issue #1066). Earlier versions used 89.9 fF and $E_C = 0.2156\,\mathrm{GHz}$. That value came from inverting the anharmonic $\omega_{01} \approx \sqrt{8E_JE_C} - E_C$ at 4.14 GHz and leaving out C_J, a misreading of a linear eigenfrequency; the tracker records the ruling to use the published value. The source spread is real: the Palace tutorial's own run of the same device gives 4.0991 GHz, which would put the target near 95.95 fF, and that run was not adopted. Two further corrections to the target, for the hybridization of the 4.14 GHz mode with the resonator and for a series loop inductance, are estimates and are reported in Section 9.5. Every committed artifact aimed at 89.9 fF carries a superseded-anchor header, and its runs are kept as history.

The gap to the design anchor was discretization error. The coarse $C_\Sigma = 136.7\,\mathrm{fF}$ *exceeds* the anchor. Earlier versions, following the benchmark as then committed, diagnosed the gap as a model-definition difference and not a truncation artifact, on two checks: grounding the exterior wall moves C_Σ by a relative 6×10^{-5}, and the scalar-versus-tensor sapphire treatment shifts it 0.75% on the coarse mesh. Neither check tests mesh convergence, and the diagnosis is withdrawn. The convergence study settles the question. The converged scalar-ε C_Σ of X^0 is certified in $[77.40, 77.61]\,\mathrm{fF}$, which is at least 16.34 fF, or 17.3%, *below* the 93.95 fF anchor; with the tensor it is below 89.50 fF, at least 4.7% below. The coarse mesh overstates C_Σ by at least 54.69 fF, more than the whole 43.75 fF by which the coarse scalar value exceeds the anchor. The gap was discretization error, and more than all of it.

What remains is a gap of the opposite sign: converged, geode-fem's electrostatic C_Σ of the unmodified geometry is below the anchor. Earlier versions suggested, as an inference, that the junction's 5.5 fF closes part of it. That inference is withdrawn: C_J is excluded from both sides of the comparison and added to both in E_C. We report the measured values and the discrepancy; we do not retrofit to the anchor. Section 9 uses the anchor as an optimization target and reports what that does and does not establish.

8 Differentiable Sensitivities: The Discrete-Adjoint Layer

8.1 The 2×2 sensitivity matrix

For a solve $A(\varepsilon, X)\,x = b(X)$ and a scalar observable $g(x)$, the discrete adjoint $A^{\mathsf{T}}\lambda = \partial g/\partial x$ turns the parameter gradient into explicit contractions:

$$\frac{dg}{d\varepsilon_k} = -\lambda^{\mathsf{T}}\frac{\partial A}{\partial \varepsilon_k}x, \qquad \frac{dg}{dX} = \lambda^{\mathsf{T}}\frac{\partial b}{\partial X} - \lambda^{\mathsf{T}}\frac{\partial A}{\partial X}x \tag{4}$$

(real case; the complex driven case takes $2\,\mathrm{Re}[\cdot]$ of the same contractions with the *transpose*, not conjugate-transpose, solve — the pencil is complex-symmetric, so the adjoint solve reuses the forward LU factorization). Everything on the right-hand side is assembly-level and therefore differentiable on the tensor substrate: the element Jacobians $\partial A/\partial X$ and $\partial b/\partial X$ are *exact*, computed by forward-mode dual numbers through the same closed-form element kernels the solver assembles — no finite-difference truncation enters the gradient. Each gradient costs one forward plus one adjoint solve sharing a single factorization (asserted programmatically: `n_factorizations` $= 1$).

Four adjoint modules are merged and validated, spanning material and geometry sensitivities on both operators (Table 4). Each carries a committed test that the analytic gradient match a central finite difference *of the full independent pipeline* (perturb the parameter, re-assemble, re-solve through the public solver path, recompute g) — and, load-bearingly, mutation tests proving the

Table 4: The merged 2×2 discrete-adjoint sensitivity matrix. Each cell: one forward + one adjoint solve sharing a single factorization; element Jacobians exact (forward-mode duals through the production kernels); validated against central finite differences of the full pipeline, with mutation tests. "FD rel-err" is the observed agreement of the analytic gradient with the FD reference in the merged test suite (test acceptance bounds are looser; see the module sources). Modules in `crates/geode-core/src/`.

Operator	Parameter	Module (issue / PR)	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\varepsilon$	`adjoint` (#570 / #573)	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	`shape` (#571 / #575)	$\sim 1 \times 10^{-9}$
driven $H(\mathrm{curl})$	material $\partial/\partial\varepsilon$	`driven::adjoint` (#576 / #579)	2.3×10^{-5} (worst region)
driven $H(\mathrm{curl})$	geometry $\partial/\partial X$	`driven::shape` (#577 / #581)	2.2×10^{-9}, 1.2×10^{-8} (two maps)

tolerances bite: the classic complex-adjoint conjugation error (A^{H} for A^{T}) is rejected by the FD check, as is a dropped term. The driven-shape case surfaced a physics subtlety worth recording: the current-source right-hand side $b = \mathrm{i}\omega\mu_0 \int N \cdot J$ is geometry-dependent, so the shape gradient carries a $\partial b/\partial X$ term absent from the material case — omitting it fails the finite-difference cross-check outright.

The matrix is deliberately 2×2: material and geometry are the two design-parameter classes, and the scalar and driven operators are the electrostatic (LOM) and RF (S-parameter/EPR-workhorse) branches of the solver. The coarse-mesh LOM tiers of Section 9 exercise the scalar column end-to-end, and the refined optimization of Section 9.4 uses the same stationarity identity in a native-f64 P2 implementation; the driven column is merged and validated at the operator level and composes with the driven solve, the one solve geode-fem's GPU path runs (correctly, but slower than both assembled CPU solvers at every measured size, and at best at parity with the same matrix-free algorithm on the CPU; Section 11.3) — RF shape optimization is the natural follow-on, not exercised here.

8.2 The capacitance$\to E_C$ chain, and a vanishing adjoint

Composing the shape adjoint with the LOM chain of Section 7 gives $\partial(C, E_C, \alpha)/\partial(\text{geometry})$ (merged as `shape::capacitance_shape_gradient`, #583 / PR #586). The composition carries an elegant simplification. The capacitance observable is the stored field energy, $C = 2W = \phi^{\mathsf{T}} K(X) \phi$, and W is *variationally stationary* in the solved potential ϕ: at the forward solution the free rows of the cotangent $K\phi$ vanish (equilibrium), so the adjoint solution $\lambda \approx 0$ and the entire shape derivative collapses to the explicit-geometry term

$$\frac{dC}{dX} \;=\; \phi^{\mathsf{T}} \frac{\partial K}{\partial X}\, \phi \qquad \text{(at the solution)}, \tag{5}$$

a Hellmann–Feynman-like result: no adjoint back-solve is needed for the capacitance itself. The implementation keeps the general adjoint form (the mutation test measures the implicit part at $< 10^{-8}$ of the total, confirming the collapse numerically rather than assuming it), and chains through $\partial E_C/\partial C_\Sigma = -e^2/(2C_\Sigma^2 h)$ exactly. Validation is double: against the analytic parallel-plate law ($\partial C/\partial d = -\varepsilon_0\varepsilon_r$ per unit area under a gap scaling; $\partial C/\partial A = +2\varepsilon_0\varepsilon_r$ under an area scaling — opposite signs, both matched) and against central finite differences of the full pipeline, with observed relative errors at the 10^{-9} level.

9 Gradient-Based Optimization: A Loop Proof, a Coarse-Mesh Detour, and a Refinement-Validated Design

Four tiers, labeled by what each one establishes: a closed-loop proof on an analytic fixture (Section 9.1); single-parameter runs on the coarse committed device mesh, which validate the gradient and are otherwise historical (Section 9.2); a three-parameter optimizer on that coarse mesh, kept as a cautionary precursor (Section 9.3); and the real-device result, a three-parameter optimization on an adaptively refined mesh with quadratic elements, followed by a held-out refinement series that brackets what the design is worth when converged (Section 9.4). Section 9.5 then asks which target the design should have been aimed at. The first two tiers use damped Newton on $E_C(\theta) - E_C^{\text{target}}$ (equivalently $C(\theta)$) with the analytic $dE_C/d\theta$ from Section 8.2; each Newton step costs one forward plus one adjoint solve on a single factorization.

One distinction governs the device tiers. The convergence study of Section 7 shows that the coarse committed mesh overstates C_Σ by at least 54.69 fF, and that the converged device lies *below* the anchor, not above it. The coarse-mesh tiers were run before that was known, and against the superseded 89.9 fF anchor. Their gradient validations and convergence statements are exact statements about the coarse discrete problem: the finite-difference checks test the derivative of the discrete C_Σ, and the fresh solves test self-consistency on one discretization. None of them is evidence about the converged device, and the direction they moved in, lowering C_Σ toward the anchor, is the wrong one for it. Only Section 9.4 measures what happens to a design under refinement. The framing claim is capability and step-count, not wall clock: a derivative-free loop (the parameter sweeps of current practice) spends N_{params} extra forward solves per step to estimate what the adjoint delivers exactly for one. No timing comparison against HFSS/Palace-based optimizers is measured or claimed.

9.1 Loop proof: Newton to a target E_C on a parallel-plate fixture

On a parallel-plate capacitor fixture ($\varepsilon_r = 3$; 343 nodes, 1,296 tets; θ scales the gap, so $d = L(1+\theta)$), Newton iteration drives the design from $C_0 = 120.0\,\text{fF}$ ($E_C = 0.1614\,\text{GHz}$) to the target $E_C = 0.2156\,\text{GHz}$, the superseded anchor value of Section 7; the loop proof does not depend on which target it is given. Full Newton converges in *one* step — and we state the reason rather than dress it up: $C \propto 1/(1+\theta)$ makes $E_C(\theta)$ *affine* in θ, so one-step convergence is expected, and this tier proves the loop end-to-end (gradient, chain rule, update, convergence test), not a hard optimization. The converged $\theta = 0.335657909$ matches the closed-form optimum to 3×10^{-15}. The honesty check that matters: a *fresh, independent* forward solve at the converged geometry — assemble, solve, extract, no adjoint machinery — reproduces the target E_C to a relative error of 1.4×10^{-15}. A damped run ($\alpha = 0.5$) converges in 13 steps and supplies the descent illustration (Figure 9a, Appendix A). At the hit target, the Koch-exact spectrum gives $E_J/E_C = 51.0$, $\omega_{01} = 4.128\,\text{GHz}$, $\alpha = -0.247\,\text{GHz}$; that ω_{01} lands near 4.14 GHz only because the superseded anchor was derived by inverting the same anharmonic formula. All trajectory values are committed in `benchmarks/transmon_diffopt/results.toml`, under a superseded-anchor header.

9.2 The coarse device mesh: validated gradients and a shortfall that was a mesh artifact

The second tier runs on the coarse committed 133k-tet SingleTransmon mesh, with a fixed-topology shape parameter θ that scales the island conductor in-plane about its centroid (101 island nodes move; the mesh connectivity is fixed). Two results stand as statements about the coarse discrete problem, and a third is historical (`benchmarks/transmon_diffopt/pad_results.toml`):

The gradient is real on the device mesh. At $\theta = 0$ the adjoint gradient $\partial C_\Sigma/\partial\theta = 198.198\,\text{fF}$ is validated against a central finite difference of the *full independent pipeline* (move the island nodes, re-assemble, re-solve, re-extract): relative error 1.15×10^{-4} at $h = 10^{-4}$, with a clean $O(h^2)$ truncation sweep ($1.2 \times 10^{-2} \to 1.0 \times 10^{-3} \to 1.15 \times 10^{-4} \to 1.0 \times 10^{-5}$ as h falls $10^{-3} \to 3 \times 10^{-5}$). This was the first real-geometry validation: the same machinery that passed at 10^{-9}-level on analytic fixtures holds up on production geometry. It is a statement about the coarse discrete problem, and the refined-mesh checks of Section 9.4 supersede it as the load-bearing one.

The loop genuinely converges on the device mesh. A demonstration target within the mesh's validity budget ($C_\Sigma = 137.0\,\text{fF}$) converges in two Newton steps — genuinely non-affine now (the gradient changes by ∼21% along the path) — and the converged design is confirmed by a fresh, independent multi-conductor extraction at relative error 5.6×10^{-6} (Figure 9b). The entire coarse-mesh run (validation sweep plus both optimization runs) takes 10.7 s.

Historical: the anchor "shortfall" was an artifact of an unconverged mesh. On the coarse mesh the unmodified device has $C_\Sigma = 137.7\,\text{fF}$, so the then-current anchor (89.9 fF, now superseded; $E_C = 0.2155\,\text{GHz}$ — the last digit differs from the 0.2156 GHz quoted elsewhere because this run targets the rounded 89.9 fF rather than the back-solved 89.843 fF; both values are artifact-faithful) looked like a reduction of 47.8 fF. Two single-parameter attempts at it stalled on mesh validity, and earlier versions reported them as the paper's honest negative.

Rigid island scale. The $\theta = 0$ gradient puts the anchor at $\theta \approx -0.241$. The fixed-topology map inverts its first tetrahedron at $\theta = -0.0097$, and the run's conservative budget (minimum tet-volume ratio 0.25) caps $\theta_{\text{safe}} = -0.0073$, where the bounded Newton run stalls at $C_\Sigma = 136.5\,\text{fF}$. By a local linear estimate that is 33× short.

Harmonic mesh motion. Extending the prescribed island velocity into the volume with a P1 Laplace solve spreads the deformation. The field D is computed once (0.34 s; the whole run takes 7.8 s), giving $X(\theta) = X_0 + \theta D$. At the same volume-ratio threshold the safe negative step grows from -0.007258 to -0.064609, an 8.90× extension. The initial derivative is 209.087 fF per unit θ, with a full-pipeline central-FD relative error of 1.054×10^{-4}. Bounded Newton reaches 128.792998 fF, confirmed by a fresh extraction, and stalls at the safe boundary, leaving 38.892998 fF to the anchor. The artifact's initial-slope extrapolation gives $\theta_{\text{anchor}} = -0.2286$, or 3.54× the harmonic budget. The sensitivity falls along the measured path, from 209.1 to 114.5 fF per unit θ, so *if the sensitivity does not recover beyond the measured endpoint* the total is at least 6.25× the safe budget (computed 6.256959, floored). That is a conditional extrapolation, not a measured bound.

These are correct statements about the coarse discrete problem, and the mesh-validity limit of a fixed-topology morph is real. The shortfall was not. It was an artifact of an unconverged mesh: converged, the unmodified device is already below the anchor (Section 7), so the 47.8 fF reduction these runs attempted is not one the device needs, and the 33× and 6.25× factors describe no physical design problem. The refined optimization of Section 9.4 moves the other way, by about 15.6 fF, and stays well inside the same volume-ratio floor. Both coarse runs use scalar trace-averaged sapphire.

9.3 A cautionary precursor: three parameters on the coarse mesh

Before the convergence study, a three-parameter optimizer was run on the coarse mesh (`benchmarks/transmon_diffopt/multiparam_gradient.toml`, `benchmarks/transmon_diffopt/multiparam_results.toml`). Three harmonic morph fields change the island length, the pad width and the cutout gap; each is exactly zero on the junction nodes and the island leads, so no parameter moves

the junction region. Their gradients agree with full-pipeline finite differences to relative errors of 8.0×10^{-7}, 3.4×10^{-10} and 1.2×10^{-8} (rounded up). A bounded Gauss–Newton optimizer with per-step volume-stiffened re-morphing then drove the coarse-mesh scalar-ε C_Σ from 137.706752 down to the superseded 89.900001 fF in 18 steps, within a 10 kHz tolerance on E_C and with fresh-solve agreement of 1.39×10^{-14}. The stiffening exponent was selected post hoc, and a neighboring exponent reaches a different geometry in 14 steps, so the design is not unique.

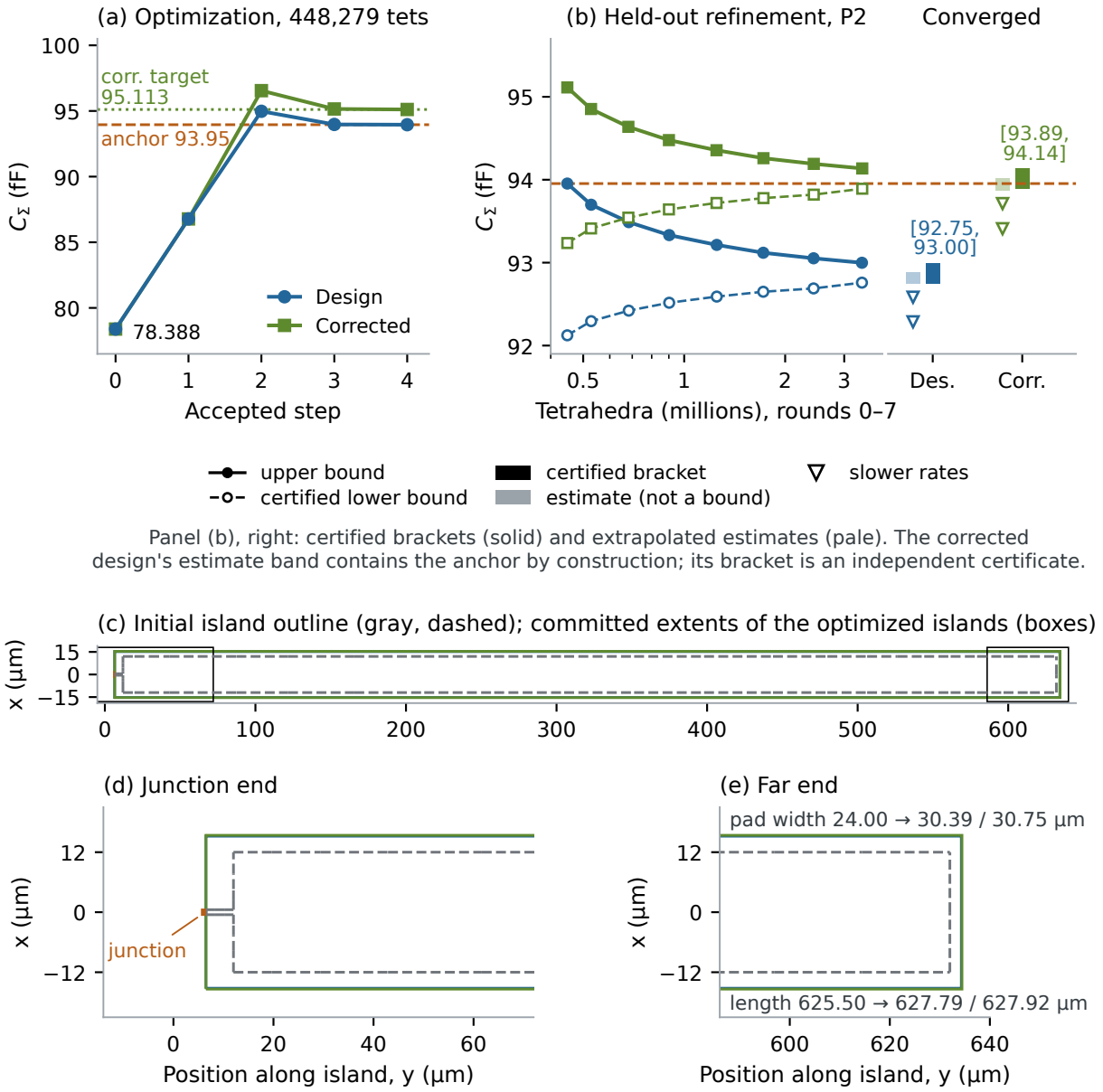
Every one of those checks passed, and as a device design the result is wrong: *the optimizer moved the wrong way.* It lowered C_Σ by 47.8 fF, whereas the converged device starts below either anchor and needs an increase. Under refinement the capacitance of that design falls to 66.609133 fF (P1, $h/2$), 56.298812 fF (P1, $h/4$) and 52.368047 fF (P2, $h/2$) (`benchmarks/transmon_diffopt/c_sigma_convergence.toml`). Its converged value is therefore below 52.37 fF, at least 44.2% below the current anchor, with an extrapolated estimate of 45.98–48.11 fF. The lesson is methodological. Finite-difference checks and fresh solves certify consistency on one discretization and say nothing about discretization error. What survives from this tier is its machinery: the junction-pinned morph fields, which the refined optimization reuses.

9.4 The real device: an optimization with a certified bracket

This is the paper's real-device design result. Its scope is the one the committed artifacts state (`benchmarks/transmon_diffopt/refined_optimization/`): the scalar-ε electrostatic island capacitance with a neutral floating feedline, on the polygonal geometry of the committed mesh, with the junction's 5.5 fF self-capacitance excluded. It is not an optimization of the full-wave eigenfrequency or of the nonlinear Hamiltonian. The target is the 93.9537 fF anchor of Section 7 (issue #1066).

Method. Ten cycles of adaptive refinement, marked by a flux-jump indicator and solved with P1 elements, turn the committed mesh into a 448,279-tetrahedron mesh. The optimizer then works on that mesh with P2 fields, solved by conjugate gradients with an algebraic-multigrid preconditioner. The geometry map is $X(\theta) = X^0 + \sum_i \theta_i D_i$, with the three junction-pinned harmonic fields of Section 9.3 (island length, pad width, cutout gap) computed on the coarse mesh and carried to each refined mesh by exact nodal prolongation. Connectivity, conductor membership and material values stay fixed, and refinement happens outside the differentiable map. The gradient is an equilibrium-energy shape derivative. For a unit island excitation the floating feedline sits at $-C_{fi}/C_{ff}$, the energy of the effective field $\phi_i - (C_{fi}/C_{ff})\,\phi_f$ is C_Σ, and stationarity removes the implicit field derivatives, as in Eq. (5). For a morph velocity v the P2 derivative integrates $\nabla\cdot v\,|\nabla\phi|^2 - 2\,\nabla\phi^{\mathsf{T}}\,\nabla v\,\nabla\phi$ with exact four-point quadrature on each affine tetrahedron. This numerical path is native f64 and does not run on the Burn tape. Accepted steps must reduce the freshly solved capacitance residual and keep the minimum element-volume ratio above 0.25. The method is that of an earlier run aimed at the superseded 89.9 fF (issue #1056), unchanged except for the target.

Optimization. On the adapted mesh the unmodified device has $C_\Sigma = 78.387918$ fF, below the anchor. The optimizer therefore *increases* C_Σ, and reaches 93.953668 fF in four accepted steps (Figure 6a; `benchmarks/transmon_diffopt/refined_optimization/p2_cycles10_anchor_93p95/summary.toml`). The final geometry has island length 627.794327 μm and pad width 30.394071 μm, from 625.5 and 24 μm (Figure 6c–e). The minimum volume ratio over the accepted steps is 0.47801. At the final design all three sensitivities are checked against central differences of two fully re-solved perturbed geometries; the worst relative discrepancy is 4.16×10^{-7} (rounded up). The target is reached on that discretization, and only there.



Panel (b), right: certified brackets (solid) and extrapolated estimates (pale). The corrected design's estimate band contains the anchor by construction; its bracket is an independent certificate.

Figure 6: The refined real-device optimization of Section 9.4: the scalar-permittivity electrostatic C_Σ (neutral floating feedline, junction's 5.5 fF excluded) from P2 solves on the committed polygonal geometry. (a) Freshly solved C_Σ after each accepted step on the 448,279-tetrahedron optimization mesh, for the design run (target: the 93.954 fF anchor) and the resolution-corrected run (target 95.112986 fF). A target reached here is reached on this discretization only. (b) The seven held-out rounds of each final design (Table 5): conforming upper bounds (filled) and certified equilibrated-flux lower bounds (open) on the same meshes. At the right, the certified brackets of the converged values, $[92.75, 93.00]$ fF and $[93.89, 94.14]$ fF; extrapolated estimates, not bounds (92.74–92.88 fF and 93.87–94.02 fF); and the same fit at two slower assumed rates, which the brackets rule out. The corrected design's estimate band contains the anchor by construction, because its series is the first one rescaled; its bracket is an independent certificate. (c) The initial island outline, from the committed input mesh, and the committed bounding boxes of the two optimized islands, at equal axis scales; (d) and (e) enlarge the two ends. The optimized outlines are not committed (Appendix A). Sources: the `summary.toml` and `design.json` files of the two runs and `c_sigma_bracket_anchor_93p95.toml`.

Table 5: Held-out refinement of the two designs aimed at the 93.9537 fF anchor. Round 0 is the optimization mesh; no later round feeds the optimizer. "Design" is the run targeted at the anchor; X^0 is the unmodified device on the same meshes, and the increment is their difference. "Corrected" is the resolution-corrected design on its own meshes. All capacitances are in fF with P2 elements, and each is an upper bound on the converged value of its geometry; the certified lower bounds on the same meshes are in the bracket artifact. Sources: the two `summary.toml` files under `benchmarks/transmon_diffopt/refined_optimization/p2_cycles10_anchor_93p95*/`.

Round	Tetrahedra	Design	X^0	Increment	Tetrahedra	Corrected
0	448,279	93.953668	78.387918	15.565749	448,279	95.113016
1	527,959	93.698450	78.178675	15.519776	527,565	94.852766
2	683,440	93.488379	78.011681	15.476698	681,721	94.637932
3	902,171	93.332106	77.882135	15.449971	900,507	94.477742
4	1,250,525	93.214010	77.784538	15.429473	1,248,566	94.355732
5	1,720,753	93.119307	77.707882	15.411425	1,723,255	94.259623
6	2,430,808	93.052517	77.651594	15.400924	2,433,998	94.191268
7	3,388,197	92.998560	77.607978	15.390582	3,395,664	94.136103

Held-out refinement: the upper side. The final geometry is then refined for seven further rounds with the same marking rule, with a fresh P2 solve each round. None of it feeds the optimizer. Table 5 and Figure 6b give the series. The capacitance falls monotonically from 93.953668 fF at 448,279 tetrahedra to 92.998560 fF at 3,388,197, and it is still falling, by 0.05 fF in the last round. The held-out spaces are nested and conforming and the capacitance is an energy minimum, so every value is an upper bound on the converged capacitance of this polygonal geometry: the converged value is below 93.00 fF, at least 1.01% below the anchor.

Held-out refinement: the certified lower side. On every held-out mesh of both designs, the equilibrated-flux certificate of Section 7 gives a guaranteed lower bound (issue #1075, PR #1077; `benchmarks/transmon_diffopt/c_sigma_bracket_anchor_93p95.toml`). The best lower bound rises from 92.123856 fF on the optimization mesh to 92.758828 fF at round 7, where a third-order flux sets it. Refinement does not move the polygonal surfaces, so every mesh bounds the same converged value, and the family bracket is the largest lower bound and the smallest upper bound, rounded outward:

$$\textit{the converged } C_\Sigma \textit{ of the design lies in } [92.75, 93.00]\,\textit{fF}, \text{ a bracket } 0.25\,\text{fF } (0.27\%) \text{ wide,}$$

which is 1.01–1.28% below the anchor, and below it by at least 0.95 fF. At round 7 the bound rises 0.10 fF from the first- to the second-order flux and 0.03 fF from the second- to the third-order flux, so what limits the width is the P2 field, not the flux order. This settles what earlier versions could not: the design is certified to sit more than 1% below the anchor it was aimed at, and the design runs show that a target reached on the optimization mesh is lost by about that much under refinement.

Estimates, kept as context. The decrement ratios show no trend toward zero. A geometric tail at the largest and smallest observed ratio gives 92.747491 to 92.869454 fF, and least-squares fits of $C_\infty + aN^{-\alpha}$ with a free rate on three windows of rounds give 92.879735, 92.846467 and 92.867953 fF; the outward-rounded union is 92.74–92.88 fF. At slower assumed rates the same fit gives 92.286273 fF ($\alpha = 1/3$) and 92.583024 fF ($\alpha = 1/2$). The certified bracket rules out both slower-rate values and the lower geometric-tail value; the estimate was never a bound, and the bracket now shows where it was wrong.

The design increment. On the same meshes the unmodified device goes from 78.387918 to 77.607978 fF. The increment the optimizer added moves much less than either capacitance: it is 15.565749 fF on the optimization mesh and 15.390582 fF after seven rounds, a change of about 1.1%. Its certified value, the design bracket minus the X^0 bracket, lies in $[15.14, 15.59]$ fF. Most of the discretization error is common to the two geometries on these meshes, which are adapted to the final design and not to X^0; that is an observation about this series, not a bound.

A resolution-corrected design. A second run repeats the optimization on the same discretization with the target offset by the measured bias: the anchor times the optimization-mesh value over the geometric-tail midpoint, $93.953660 \times 93.953668/92.808473 = 95.112986$ fF. The first run's artifact records that target, which was fixed before the second run (`benchmarks/transmon_diffopt/refined_optimization/p2_cycles10_anchor_93p95_resolution_corrected/summary.toml`). The optimizer reaches 95.113016 fF in four accepted steps (island length 627.924937 μm, pad width 30.749537 μm, minimum volume ratio 0.43912, worst finite-difference discrepancy 4.24×10^{-7}). Its held-out series runs from 95.113016 to 94.136103 fF at 3,395,664 tetrahedra, and its certified lower bound rises to 93.891607 fF. The converged capacitance of this geometry therefore lies in $[93.89, 94.14]$ fF, which contains the anchor: the converged value is between 0.07% below and 0.20% above it.

What is by construction and what is not. The corrected series is the first series rescaled: round by round, the ratio of the two is constant to a relative spread of 1.1×10^{-4} (largest ratio over smallest, minus one; our computation from the two committed series). An extrapolation rule applied to both therefore returns, to about that accuracy, the first run's extrapolated value times that ratio, and the target was set to the anchor times the inverse of the first run's extrapolated bias. **The estimate band of 93.87–94.02 fF contains the anchor by construction.** The bracket does not: its lower bound is a certificate computed from the second design's own fields, and it could have excluded the anchor. That it does not is the measurement. The corrected design's slower-rate extrapolations, 93.409179 and 93.712148 fF, lie below its certified lower bound and are ruled out. The two runs share a mesh family, a marking rule and an element order, so the bias transfer between them is not an independent confirmation; the certificate is what makes the second design's position known.

Cross-solver check: the same discretization. Palace, run with second-order elements on the exported adapted mesh of the design, gives 93.953434 fF against geode-fem's 93.953668 fF, a relative difference of 2.5×10^{-6}. On the resolution-corrected design it gives 95.112780 against 95.113016 fF, the same relative difference. Palace adds 3,777 elements by local bisection at the interior boundaries, so the two codes share the polygonal geometry and coefficients, not one mesh. This shows that two solvers agree on the same discretization. It is not evidence about the discretization error, which the held-out series and the certificate measure. No higher-order or adaptive Palace run was made at this anchor.

Superseded runs. The same method was first run against the mis-derived 89.9 fF anchor (issue #1056; `p2_cycles10/` and `p2_cycles10_resolution_corrected/` in the same directory). Those runs reached 89.900518 and 90.980104 fF in three accepted steps each, and the certificate brackets their converged values in $[88.78, 89.02]$ and $[89.84, 90.08]$ fF (`benchmarks/transmon_diffopt/c_sigma_bracket.toml`). Measured against 89.9 fF they reproduce the pattern above, one design about 1% low and the corrected one containing its target. They are history: their target is wrong, and they say nothing about the current one. Their held-out series supply the X^0 bracket, which does not depend on any anchor.

What is not claimed. No scalar-to-tensor equivalence, layout regeneration or re-meshing, full-wave optimization, experimental validation, global optimum or speedup against Palace is claimed. The certified brackets are statements about the converged electrostatic C_Σ of each polygonal geometry, not about a physical device; the geometry has zero-thickness sheet conductors and a natural zero-charge outer boundary. The choice of P2 elements and ten refinement cycles was made after exploratory P1 runs, which are retained in the artifact directory; it was not preregistered.

Roadmap. Two extensions are planned and not built. The first is a re-optimization at the loop-corrected target of Section 9.5. The second is a second design target, the island–claw coupling ratio $\beta = C_g/C_\Sigma$ (issue #1037): the coarse-mesh optimizer interface accepts several targets and is unit-tested with two targets and three parameters, but no two-target experiment on the device exists.

9.5 Which target? The anchor chain

The bracket of the resolution-corrected design contains the anchor, and the bracket is 0.25 fF wide. Whether the design is *right* then depends on whether the anchor is. The anchor reads a hybridized eigenmode of a circuit with more in it than L_J, C_J and C_Σ as if it were that bare three-element circuit. Two measurements on the committed fixture correct it, each in a direction the physics fixes and by an amount that is not converged (Figure 7).

Step 1: remove the qubit–resonator hybridization. DeviceLayout's 4.14 GHz is the lower of two coupled modes, while C_Σ treats the resonator as ground. Level repulsion from the 5.591 GHz mode pushes the reported qubit down, so the bare-qubit target is below 93.9537 fF. The coupling is measured by an avoided-crossing sweep (issue #1078, `benchmarks/transmon_coupling/results.toml`): 27 junction inductances from 4 to 30 nH with the rank-1 shunt of Section 6.3, crossing near 6.6 nH with a smallest sampled splitting of 92.15 MHz. A two-node Maxwell-form circuit fits all 54 branch frequencies with 6 kHz RMS and gives $k = 0.017866$, $g/2\pi = 46.02$ MHz at the crossing; Palace on the same mesh gives k in a ratio of 0.99997 to it. The coupling is not converged: Palace on one uniform refinement gives $k = 0.022798$ ($g/2\pi = 61.70$ MHz, 28% higher), and an electrostatic proxy keeps rising to $h/4$. Four extrapolations put the converged k at 0.02773–0.02991; the anchor uses $k = 0.0288$ with a band of 0.0227–0.0310, *an extrapolation, not a bound*. Inverting the dressed pair at that k gives a bare qubit at 4.14209 GHz and a target of 93.853 fF, band 93.826–93.892 fF; the band also carries the resonator's odd harmonics.

Step 2: add the series loop inductance. Section 6.1 identified a loop inductance L_s in series with the junction; reading 4.14 GHz through L_J alone ignores it, and including it lowers the target further. L_s is the geometric inductance of the qubit's current path, the junction wire and the 620 μm island finger with its ground return, and it is not mesh-converged (issue #1079, `benchmarks/transmon_loop_inductance/results.toml`): Palace gives 0.0852 nH on the fixture, 0.1047 nH on its own uniform refinement and 0.1132 nH at second order on the fixture. An $h/4$ solve, about 11 million unknowns, does not fit on the local host and was not run. Richardson extrapolation at order one, the slowest rate the sheet-edge theory allows, gives 0.1242 nH, and at the static capacitance's observed order 0.1164 nH; the anchor takes their midpoint, $L_s = 0.120$ nH, with a band of 0.113–0.125 nH from the finest measured value to the order-one estimate, *an extrapolation, not a bound*. A coplanar-line formula for the finger plus the wire gives 0.117 nH, but it assumes the charge is spread uniformly along the finger and is an order-of-magnitude check only; its closeness

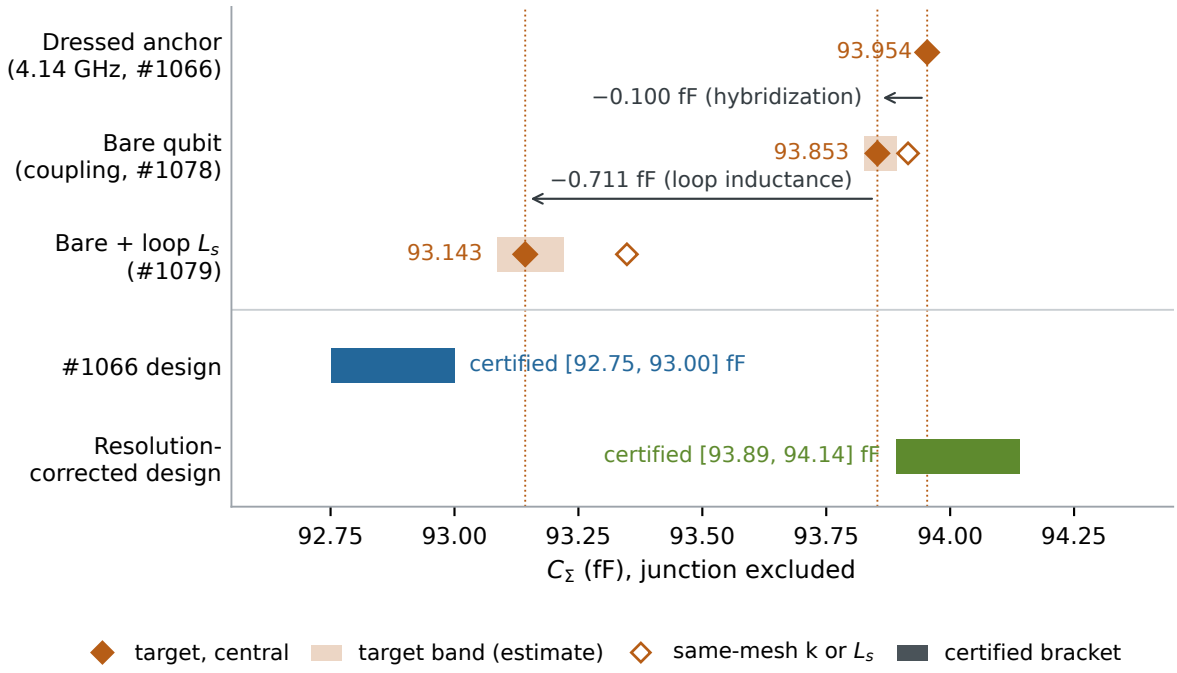


Figure 7: The anchor chain against the two certified design brackets. Top: the dressed anchor (93.954 fF, the published 4.14 GHz read as a harmonic frequency), the bare-qubit target after removing the measured hybridization (93.853 fF) and the target with the series loop inductance added (93.143 fF). The shaded bands are estimates from the extrapolated coupling and loop inductance, not bounds; the open markers are the same targets with the coarse fixture's own k or L_s, the alternative that the transfer assumption names. Bottom: the certified brackets of the converged C_Σ of the two designs, rounded outward. Neither bracket contains the most corrected target. Sources: `benchmarks/transmon_coupling/results.toml`, `benchmarks/transmon_loop_inductance/results.toml` and `benchmarks/transmon_diffopt/c_sigma_bracket_anchor_93p95.toml`.

to 0.120 nH is not corroboration. Read through the series circuit, the bare qubit gives a target of 93.143 fF, band 93.087–93.222 fF, 0.711 fF below the bare target.

The transfer assumption. Both corrections assume that one k and one L_s, measured on geodefem's meshes and extrapolated to convergence, describe the run that produced DeviceLayout's 4.14 GHz. That run used its own mesh, and it is not reproduced here. If it was as coarse as the committed fixture, the fixture's own k and L_s apply instead, and the two targets would be 93.915 and 93.348 fF (the open markers of Figure 7).

Verdicts. Table 6 sets the brackets against the targets. Against the dressed anchor, the corrected design is in and the first design is out. Against the bare target the corrected design is above it, by 0.0002 fF at the edge of the target band, so that edge is not resolved. Against the loop-corrected target *neither* design brackets it: the first is below by at least 0.088 fF over the whole band, and the corrected one above by at least 0.669 fF. The two designs straddle the most corrected target, and a re-optimization aimed at 93.143 fF is the obvious next step; it has not been run.

Where the uncertainty is. The optimizer converges to whatever target it is given, and the certified bracket, 0.25 fF wide, decides design against target. The two anchor corrections move the

Table 6: The two certified design brackets against the three targets. Each entry is the bracket minus the central target, rounded outward, in fF; the last column is the verdict over the whole target band. The brackets are certified; the bare and loop-corrected targets are estimates resting on extrapolated k and L_s and on the transfer assumption. Sources: `c_sigma_bracket_anchor_93p95.toml`, and the `[verdicts]` of `benchmarks/transmon_coupling/results.toml` and `benchmarks/transmon_loop_inductance/results.toml`.

Design (certified)	Dressed 93.954	Bare 93.853	Bare + loop 93.143	Over the bands
Design [92.75, 93.00]	−1.20 to −0.95	−1.10 to −0.85	−0.39 to −0.14	below all three
Corrected [93.89, 94.14]	−0.07 to +0.19	+0.03 to +0.29	+0.74 to +1.00	contains dressed only

target by 0.100 and 0.711 fF, and the loop-corrected band alone is 0.135 fF wide; the same-mesh alternatives differ from the central targets by up to 0.21 fF. The remaining uncertainty of this design problem is therefore in the physical anchor — in the unconverged k and L_s and in the transfer assumption — and not in the solver. The loop correction also closes most of the frequency-side offset of Section 6.1: on the fixture, 94% of Palace's 0.28% offset from the lumped estimate is L_s.

10 Roadmap: Differentiating the Eigenmode Branch

The LOM branch above differentiates an *electrostatic* solve. The transmon's dynamical figures-of-merit — resonator frequency, participation-ratio and EPR self/cross-Kerr (Minev et al., 2021a; Nigg et al., 2012) — live in the *eigenmode* problem, and that branch is **not differentiable yet** in geode-fem. The paper commits to the explicit split:

Branch	Observables	Differentiable?
LOM (electrostatic)	E_C, $\alpha \approx -E_C$, coupling C	**yes** (Sections 8–9)
eigenmode / EPR	resonator f, participation, Kerr	**not yet** (this section)

Differentiating an eigenpair needs the adjoint-eigenpair (Hellmann–Feynman / Nelson) formulas (Nelson, 1976), which require solving the interior eigenproblem robustly at the physical shift — and geode-fem's matrix-free interior eigensolve is blocked at exactly that shift. The blocker is measured and committed (`benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`): at $\sigma = 4.5$ GHz on the 133k-DOF fixture, a *single* inner AMS-preconditioned MINRES solve (outer Lanczos step 0) never converges — it plateaus on a nearly flat residual tail below $\sim 10^{-5}$ and was still running at 14,300 inner iterations ($\|r\| \approx 2.6 \times 10^{-6}$) when killed. The plateau is *coarse-solve-invariant* — an exact coarse solve stalls the same way — so the limiter is the SPD-proxy preconditioner $K + |\sigma|M$ itself, not the coarse solve. Nor is there a single inner tolerance that is both cheap and correct: at an inner tolerance of 10^{-2} the entire eigensolve completes in $\sim$56 s but returns the non-physical $\lambda \approx 0$ gradient near-kernel cluster (0.42–0.64 GHz), not the physical band. The named path — not built, cited as method, not result — is a preconditioned Helmholtz-projected Jacobi–Davidson iteration, applying the gradient-kernel projection inside the correction equation, for which Liang et al. (2026) provide the current adaptive-multilevel treatment of exactly this singular Maxwell eigenproblem; combined with the Nelson/Hellmann–Feynman eigenpair derivatives, that is the route to differentiable eigenmode/EPR quantities. Until it lands, no claim in this paper extends to derivatives of the eigenmode spectrum.

Table 7: Mode-verified CPU timings on one m6i.4xlarge: median wall seconds / measured core-seconds. All rows contain the six reference modes of Table 1. The solvers start from the same input mesh but Palace solves 183,891 unknowns and geode-fem 156,863; the timings are not normalized for that. Three repeats per cell except Palace one-rank Save=0 (one run). The two oracle-tuned requests are marked †; the historical port-aware route is the pre-port-subspace implementation. Palace always requests six modes above 4.5 GHz.

Request	Extra modes	1 wide	8 wide
geode 30 at 4.5 GHz†	24	34.45 / 34.44	29.08 / 40.71
geode 6 at 20 GHz†	0	33.21 / 33.20	27.26 / 38.87
geode port-aware 8 at 4.5 GHz	2	67.26 / 67.25	51.12 / 91.72
Palace, Save=6	0	186.84 / 186.82	33.35 / 266.66
Palace, Save=0	0	113.00 / 112.98	23.64 / 188.89

11 Performance Context

The timings below qualify the cost of the capability. Eigenmode, driven CPU and driven GPU results are separate experiments; none measures the optimizer's cost or establishes a general solver ranking.

11.1 CPU eigenmodes: same box, verified mode content

The earlier timing request, six modes at 4.5 GHz, did not establish equivalent work. In the mode-logged re-run it returns four near-kernel modes, the 3.45 GHz qubit mode and only one of Palace's six modes in geode-fem, whereas Palace, which searches only above its target, returns its six. The historical run has no committed mode log or CPU-time record; its mode content and actual thread count cannot be recovered. We therefore withdraw the earlier claim that one geode core beats eight Palace ranks with roughly twelve times fewer core-seconds. This does not change the separate frequency agreement credential of Section 5.

The replacement experiment ran on one otherwise idle EC2 `m6i.4xlarge` in `us-east-1` on 2026-10-08, with geode `61e8571e` and Palace `fba6a5b`. One-wide runs were OS-pinned to one CPU; eight-wide runs used one hardware thread per physical core. Wall clock covers the whole solver process; core-seconds are measured user plus system CPU time, not wall time multiplied by requested width. Table 7 and Figure 8 retain all three geode requests that recover the six reference modes. Two are tuned using the reference spectrum: requesting thirty modes at 4.5 GHz, or moving the shift to 20 GHz. The older port-aware extraction route does not use reference frequencies to choose its request, but returns two further modes alongside the six (one near-kernel mode and the qubit mode); matching the six still uses Palace. Palace's target is a lower bound, whereas geode's shift is a centre, so identical numerical requests need not select identical spectra.

Two asymmetries qualify every comparison in Table 7. First, the solvers do not solve the same number of unknowns. Palace refines the input mesh around the metal sheets, and every timed Palace run assembles 183,891 Nédélec unknowns against geode-fem's 156,863 (133,108 interior), about 17% more (Section 4.1). The table does not correct for this, and the difference favours geode-fem. Second, the reference set is Palace's six modes above 4.5 GHz. That set includes the 17.49 GHz port-sheet mode and excludes the qubit mode, which Palace cannot return at this target (Section 6). The "Extra modes" column counts returned modes outside that set. In the thirty-mode and port-aware rows one of the extras is the qubit mode, which the committed artifact classes as `qubit` since its re-baselining (it was `spurious_port`); the others are near-kernel modes. The counts themselves did not change: the artifact now calls this column `modes_outside_palace_request`.

At eight wide Palace without field output takes 23.64 s, less than any of these geode routes

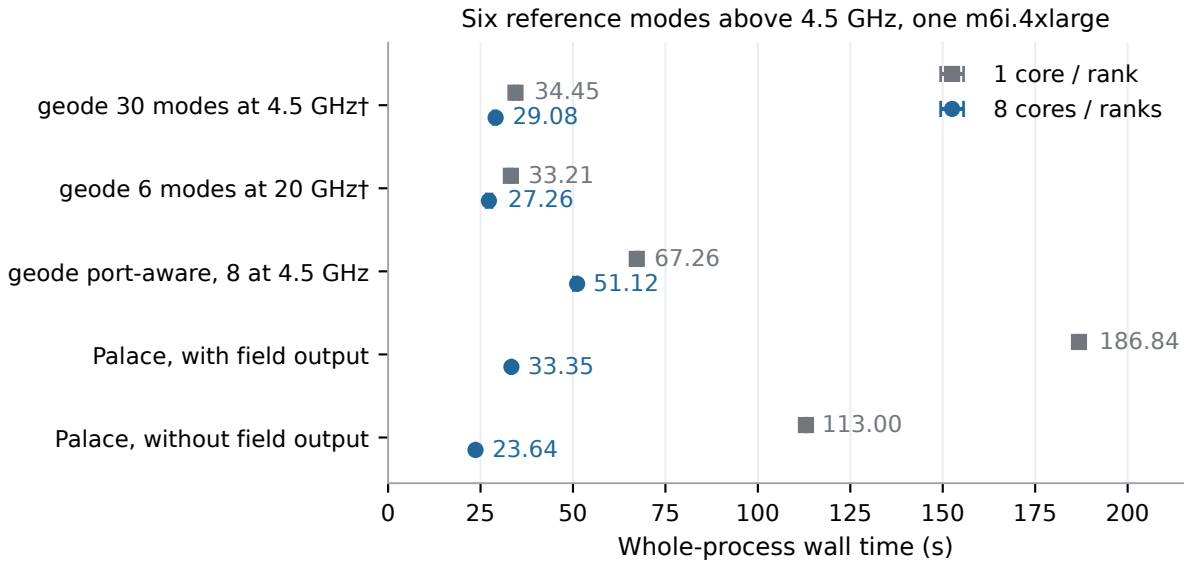


Figure 8: Whole-process wall time on one m6i.4xlarge for the three mode-recovering geode requests and for Palace with and without field output (Table 7), on one linear axis for one and eight cores or ranks. Markers are medians; whiskers span the observed minimum and maximum. Every request returns the six reference modes of Table 1, which include the 17.49 GHz port-sheet artifact and exclude the qubit mode. The two oracle-tuned requests (marked †) are not a reference-free recipe for an unknown device. Save=0 removes Palace's field output but retains its error estimator, and Palace solves about 17% more unknowns from the same input mesh, so these are workload comparisons, not solver-kernel speedups.

(27.26–51.12 s). With field output, Palace takes 33.35 s and the two oracle-tuned routes are faster. At one wide the geode routes take less wall and CPU time, but neither comparison isolates a solver kernel: geode writes no fields and has no counterpart to Palace's error estimator, which remains enabled even with Save=0 (about 40.2 s at one rank and 5.0 s at eight). MPI busy-waiting also affects measured Palace core-seconds. Palace also carries about 17% more unknowns in every cell. No single route is selected as a headline winner.

The Palace build uses SuperLU-preconditioned GMRES, not AMS: the pinned source and build configuration resolve `Default` to SuperLU. A later, separate m6i session with a tuned Palace build found only a 1.01–1.05× same-box wall-time improvement over its untuned control; cross-session differences must not be credited to compiler tuning. Those runs also predate the new port-subspace solver. The local update in Section 6.2 cannot be substituted into this same-box timing table.

At 1,157,564 DOFs, a historical geode direct run completed on a 128 GB box in 565.5 s with 92.2 GB peak RSS. Palace's earlier eight-rank run recorded 423.12 s and about 33 GB aggregate memory. These are different-host observations, and the geode request lacks mode-content verification. The DOF count is geode-fem's; we infer that Palace's internal refinement applies to that mesh as well, and its unknown count for that run is not quoted here. They establish the direct factorization's large memory requirement, not a controlled performance crossover or equivalent physical-mode throughput.

Table 8: Driven CPU-f64 linear-solve seconds on one r6i.4xlarge, one core/rank, median of two geode and three Palace runs, except 897,650 edges (one run each). geode includes operator assembly, preconditioner setup, iteration and residual check; Palace sums its Setup, Preconditioner and Linear Solve timers, excluding Operator Construction. These are not whole-process timings.

Edges	geode Jacobi	geode AMS	Palace AMS	Palace SuperLU
25,695	2.4	0.7	1.0	1.4
102,024	22.2	5.3	5.0	8.9
462,520	227.3	56.4	26.4	85.9
897,650	624.3	181.3	53.3	235.8

11.2 CPU driven solve: AMS at scale

A separate f64 experiment uses assembled COCG on the lossy driven cube at $\omega = 0.10$, on one dedicated `r6i.4xlarge`, with geode `64d44269`. AMS reduces one-core linear-solve time relative to Jacobi by 3.4–4.2× across 25,695–897,650 edges. It reduces iterations from 2969–13662 to 39–105, but does not eliminate iteration growth. Every geode solve meets an explicitly recomputed relative-residual tolerance of 10^{-8}.

At 462,520 edges AMS needs about 3.4 GB versus Jacobi's 1.05 GB. At 897,650 edges it needs 8.5 GB versus 2.0 GB, and its setup alone takes 53.488 s. Eight threads reduce setup but do not accelerate geode's Krylov iteration; at 462,520 edges its AMS solve takes 50.4 s versus Palace AMS's 4.2 s on eight ranks. The one-core geode AMS advantage at the smallest size does not persist at scale: Palace AMS takes 53.3 s against 181.3 s at the largest size. These measurements concern one structured first-order fixture, assembled CPU operators and low frequencies. They establish neither GPU performance nor a remedy for the interior eigenproblem.

11.3 GPU cell: correctness proven, scaling an honest negative

geode-fem's GPU path runs the *driven* solve (matrix-free apply + on-device COCG), not the eigensolve of the credential, and all GPU execution is CUDA-f32 (Section 3). On 2026-07-14, geode-fem code executed on a physical GPU for the first time (EC2 `g6e.xlarge`, 1× NVIDIA L40S 46 GB, driver 595.71.05, CUDA 13.2): the two CUDA-f32 correctness smokes — the matrix-free Nédélec matvec and the on-device COCG — pass against the f64 CPU reference within f32-appropriate tolerances (15.0 and 3.3 s wall, durations dominated by GPU init and kernel JIT — correctness evidence, not performance).

The scaling study (committed as `benchmarks/gpu_driven_scaling/results.toml`; a σ-lossy lumped-port cube fixture at fixed frequency, mesh-refined from 1,854 to 25,695 edges, all sixteen cells on the same host) is an honest negative: **GPU-f32 matrix-free is slower than both assembled CPU configurations at every measured size, and slower than CPU matrix-free at every size except the largest**, where the two solve-only medians are at parity (81.764 versus 82.056 s; Table 9). Computed from the artifact's unrounded medians, the assembled-CSR COCG on CPU is 136× faster at 1,854 edges (displayed: 0.032 versus 4.39 s) and still 44× faster at 25,695 (1.86 versus 81.8 s); even the direct LU is 13.5× faster at the top size (6.04 versus 81.8 s). The mechanism is kernel-launch domination (∼28 ms per iteration at the top size versus ∼0.63 ms for assembled-CSR on CPU) — 25.7k edges is far too small to fill an L40S — and f32 accuracy compounds the verdict (relative error degrading to 1.2×10^{-3} with refinement; nondeterministic convergence at the top size). One directional positive survives, and it is the one the architecture predicts: against the *same matrix-free algorithm on CPU*, the GPU's deficit falls monotonically to parity at ∼26k edges (81.8 versus 82.1 s). We state the GPU position as trajectory, not achievement, with tracked levers

Table 9: Driven-solve wall-clock scaling, solve-only medians of $n = 3$ timed repetitions (warm-up excluded), same host (`g6e.xlarge`: 4 vCPU + 1× L40S). One fixture (σ-lossy lumped-port cube) at fixed ω, mesh-refined across the columns. f64 iterative configurations at tolerance 10^{-8}; the f32 GPU configuration at 10^{-6} (its recurrence floor sits above 10^{-8}). The accuracy row is the full-field relative L_2 error of the GPU-f32 solution against the Direct-f64 reference. At 25,695 edges the GPU-f32 configuration converged in the committed single-frequency runs but not in its five-point frequency sweep, and run-to-run convergence is nondeterministic at that size (f32 floor). Values verbatim from the committed `benchmarks/gpu_driven_scaling/results.toml`.

	Mesh size (Nédélec edges)			
Configuration	1,854	5,859	13,428	25,695
Direct faer LU (CPU f64), s	0.024	0.203	1.540	6.036
Assembled-CSR COCG (CPU f64), s	0.032	0.198	0.709	1.865
Matrix-free COCG (CPU f64), s	1.652	8.885	30.234	82.056
Matrix-free COCG (GPU f32), s	4.388	13.351	34.381	81.764
GPU-f32 accuracy vs. Direct-f64 (rel. L_2)	1.26×10^{-4}	2.93×10^{-4}	4.60×10^{-4}	1.20×10^{-3}

(kernel fusion; on-device f64/mixed-precision when cubecl ships it; preconditioning) — and we prefer publishing the bound to implying the win. This cell is a cube fixture family, not the transmon eigenproblem, and its CPU baselines (4-vCPU host) are not comparable to the `m6i` numbers of Table 7.

12 Reproducibility

Every artifact behind every result in this paper is committed and scripted.[3] The exceptions are labelled where they occur: the scratch-probe measurements and source readings of Section 6 trace to a committed design note and probe source, not to a regenerable benchmark artifact.

- **Geometry and mesh.** Generated by DeviceLayout.jl v1.15.0's `SingleTransmon` example (AWS Center for Quantum Computing, 2025); the fixture generator is committed, and the MSH 4.1 fixture is pinned by SHA-256 (`5b3ff4c3...b33dd`; full hash and provenance in the committed `transmon_smoke.provenance.txt`).
- **The MSH conversion gotcha.** MFEM's MSH 4.1 reader rejects the DeviceLayout multi-entity-block file, so the Palace runs consume an MSH 2.2 conversion (`gmsh -save -format msh2`). The conversion is node-preserving (all 22,684 nodes unchanged, physical groups preserved) and physics-neutral: eigenvalues reproduce bit-for-bit across the conversion. Documented in the committed `palace_config.provenance.txt` because it is exactly the kind of silent pipeline step a same-mesh claim must disclose. A second such step went undisclosed in earlier versions: Palace refines the mesh it reads (4,913 elements added by local bisection, 6,014 duplicated vertices) and solves 183,891 unknowns, as its committed run logs record (`reference/fixtures/transmon_palace/results_p1/palace_run_v22.log`).
- **Credential artifacts.** The Palace configuration and provenance under `reference/fixtures/transmon_palace/`; geode-fem's benchmark definitions, the agreement numbers, the L-doubling tripwire (17.4901 → 12.37 GHz, committed ratio 0.7071), the qubit-mode measurements (3.4528 GHz, $p = 0.9942$; L-doubling ratio 0.7081; stored under `[modes.qubit]` and `[tripwires.qubit_l_doubling]`, `[spurious_mode]` before issue #1045), the opt-in rank-1 shunt (`[port_voltage_a2]`), Palace's `Target` = 2 GHz re-run (`[oracles.palace_target2]`),

[3] Repository URL and archival DOI for the geode-fem source to be added at submission. TODO(operator).

and the full gauge/projection arc of Section 6.2 (tree–cotree counts and the 1.64% shift; bulk-projection divergence ratio 50.2 and projected norm 1.06×10^{-4}; port-aware acceptance at $\leq 0.029\%$) in `benchmarks/transmon_eigen/results.toml`, each pinned by a release-tier regression test. Palace's raw outputs (`eig.csv`, per-port participation, run log) are committed; a rerun reproduced `eig.csv` bit-for-bit. The re-run's configurations, logs, `eig.csv`, `port-EPR.csv` and provenance are under `reference/fixtures/transmon_palace/results_p1_target2/`. The source reading of Palace's port operator and search window, the port-voltage probe and the planned rank-1 formulation are in `docs/research/transmon-lumped-port-formulation.md`; its probe is scratch source that the build does not compile.

- **Precision of the committed agreement numbers.** `benchmarks/transmon_eigen/results.toml` stores geode-fem frequencies at three decimals while the Palace column carries full precision, so the committed per-mode `rel_err_pct` is computed against the rounded geode-fem values; at full precision the agreement is slightly better (approximately 0.029% worst case). We quote the committed-artifact numbers throughout and disclose the rounding here. Derived ratios quoted in the text (core-second and speedup multipliers) are computed from the displayed table values unless marked as committed.

- **LOM chain.** The three-conductor capacitance extraction and Koch-exact spectrum on the coarse mesh ($C_\Sigma = 136.7\,\text{fF}$, $E_C = 0.136\,\text{GHz}$ with C_J, the anchor derivation in `[quantum.anchors]`, and the boundary and sapphire-model checks of Section 7) in `benchmarks/transmon_quantum/results.toml`; pinned by the release test `crates/geode-core/tests/transmon_quantum.rs`. The mesh-convergence study (P1 to $h/4$, P2 to $h/2$, scalar and tensor permittivity, with its bounds, estimates and memory record) in `benchmarks/transmon_diffopt/c_sigma_convergence.toml`, written by the example `transmon_c_sigma_convergence`; it ran locally in under five minutes; earlier runs of it peaked at 12.8–14.4 GB of resident memory. The certified lower bounds in `benchmarks/transmon_diffopt/c_sigma_bracket.toml` (X^0 and the superseded designs) and `benchmarks/transmon_diffopt/c_sigma_bracket_anchor_93p95.toml` (the current designs), written by the example `transmon_c_sigma_bracket`, which asserts that every primal value reproduces its committed artifact; the two parts took 102 and 64.5 min locally, at 35.5 and 29.6 GB peak resident memory.

- **Adjoint layer.** The four sensitivity modules with their FD-and-mutation test suites: `crates/geode-core/src/adjoint.rs` (PR #573); `crates/geode-core/src/shape.rs` (PRs #575/#586; this module also carries `capacitance_shape_gradient`); `crates/geode-core/src/driven/adjoint.rs` (PR #579); and `crates/geode-core/src/driven/shape.rs` (PR #581).

- **Optimization demonstrations.** The parallel-plate loop proof (full trajectories, closed-form check, fresh-solve confirmation) in `benchmarks/transmon_diffopt/results.toml`. The refined real-device optimization under `benchmarks/transmon_diffopt/refined_optimization/`: the current run directories `p2_cycles10_anchor_93p95/` and `p2_cycles10_anchor_93p95_resolution_corrected/`, and the superseded `p2_cycles10/` and `p2_cycles10_resolution_corrected/`. Each holds a generated `summary.toml` with the anchor derivation, the optimizer steps, the final finite-difference checks, the held-out series and every upper bound and estimate with its rounding direction, plus a `design.json` and the Palace configurations, logs and outputs; `validation_summary.json` is a hand-assembled index of the superseded runs, and the directory's `README.md` gives the

commands (the current runs took 29 and 12 minutes locally, at 24.6 and 22.5 GB peak resident memory). The coarse-mesh runs: the single-parameter FD validation and bounded anchor attempt in `benchmarks/transmon_diffopt/pad_results.toml`; harmonic motion in `benchmarks/transmon_diffopt/harmonic_results.toml`; the three-parameter gradients in `benchmarks/transmon_diffopt/multiparam_gradient.toml`; and the bounded optimizer runs in `benchmarks/transmon_diffopt/multiparam_results.toml`. Every artifact aimed at 89.9 fF carries a superseded-anchor header. All are consumed by `crates/geode-core/tests/transmon_diffopt.rs`, which pins the four `summary.toml` files and both bracket artifacts and re-derives their bounds, brackets and estimates.

- **The anchor chain.** The coupling sweep, its fits, the Palace cross-check cells, the static proxy, the extrapolation rule for k, the bare-qubit anchor and its verdicts in `benchmarks/transmon_coupling/results.toml` (example `transmon_coupling_g`, about 15 minutes locally; Palace cells under `reference/fixtures/transmon_palace/results_coupling_1078/`). The loop-inductance patch, topology and convergence tests, the extrapolation rule for L_s, the loop-corrected anchor, its verdicts and the offset decomposition in `benchmarks/transmon_loop_inductance/results.toml` (example `transmon_loop_inductance`; Palace cells under `reference/fixtures/transmon_palace/results_loop_1079/`). The three anchors are `TransmonAnchor::devicelayout`, `devicelayout_bare` and `devicelayout_bare_loop` in `crates/geode-core/src/quantum/transmon.rs`; tests in `crates/geode-core/tests/transmon_eigenmode.rs` pin both artifacts.

- **Performance artifacts.** The historical CPU cell in `benchmarks/transmon_bench_cpu/results.toml`; mode-verified timing in `benchmarks/transmon_bench_cpu/results_like_for_like_m6i.toml`; tuned-build controls in `benchmarks/transmon_bench_cpu/results_tuned_palace_m6i.toml`; local port-subspace evidence in `benchmarks/transmon_bench_cpu/results_like_for_like_local.toml`; driven CPU AMS results in `benchmarks/gpu_driven_scaling/results_ams_cpu_r6i.toml`; the 1.16M-DOF completion run log at `benchmarks/transmon_bench_cpu/geode_runs_1p16M_2026-07-15.log`; the GPU scaling cell (all sixteen cells, iteration counts, recomputed true residuals, DNF handling, same-host provenance) in `benchmarks/gpu_driven_scaling/results.toml`; the deep-shift eigensolve characterization at `benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`.

- **Hardware disclosure.** CPU cell: EC2 `m6i.4xlarge` (8 physical cores / 16 vCPU, 64 GB), `us-east-1` for the mode-verified re-timing (the historical session was in `us-west-2`); driven AMS: `r6i.4xlarge`, `us-east-1`; the 1.16M-DOF completion row: a 128 GB `r6i`-class box (see the committed log). GPU cell: EC2 `g6e.xlarge` (4 vCPU, 30 GB; 1× NVIDIA L40S 46 GB; driver 595.71.05, CUDA 13.2). The Palace `Target` = 2 GHz re-run: a local Apple M3 Ultra, Palace built from the in-repository Docker recipe (arm64), eight MPI ranks; no timing from it is used.

- **Versions.** geode-fem commit `3174015` (headline credential and historical CPU cells; the adjoint/diffopt layers are merged post-`3174015` and cited by PR number); Palace commit `fba6a5b`; DeviceLayout.jl v1.15.0; Burn/burn-cuda 0.21.

13 Discussion

What the credential does and does not establish. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) the cross-validation is code verification and code-to-code comparison, not validation against nature: two independent implementations agreeing to $\leq 0.033\%$ from the identical input

mesh, on discrete problems that differ by Palace's internal refinement, with scaling tripwires on the port operator and on the qubit mode. It does not certify that either solver matches a measured device — that is what Ye et al. (2025)-style experimental benchmarks address — but it rules out the large error class living in formulation, assembly, boundary handling, and eigensolver machinery. It establishes that geode-fem's eigenmode path reproduces a production solver from the same input. It does not establish mesh convergence. In particular the electrostatic observable that this paper differentiates is not converged on this mesh (Section 7), and the eigenmode credential says nothing about that.

What the differentiable-design evidence supports. The proven capability is bounded and we state its boundary. Proven: the 2×2 adjoint matrix at operator level (FD-validated with mutation tests); the capacitance$\to E_C$ chain with its stationarity collapse; a closed Newton loop confirmed by an independent fresh solve on an analytic fixture; a three-parameter optimization of the real device on an adaptively refined P2 discretization, with gradients validated against fully re-solved finite differences at 4.24×10^{-7} or better; and certified two-sided brackets on the converged capacitance of each optimized design ($[92.75, 93.00]$ fF and $[93.89, 94.14]$ fF; Section 9.4) and of the unmodified device ($[77.40, 77.61]$ fF). The first design is certified to lie more than 1% below the anchor it was aimed at, and the second to contain it. Estimated, not proven: the bare and loop-corrected targets (93.853 and 93.143 fF), which rest on an extrapolated coupling and loop inductance and on a transfer assumption; the extrapolated convergence bands, which the brackets supersede; and the corrected design's estimate band, which contains the anchor by construction. Not proven, and not claimed: that either design meets the loop-corrected target (neither bracket contains it); eigenmode/EPR derivatives (the Section 10 wall); two-target or coupling-constrained optimization; tensor-ε shape chains; a design regenerated by the layout toolchain or re-meshed; and any wall-clock superiority over sweep-based workflows. The capability is also not qubit-specific — shape and material sensitivities of electrostatic and driven-EM observables apply to RF and photonic device design generally; the transmon is the demonstration vehicle with the clearest documented unmet need.

Threats to validity. A common-mode error — both solvers wrong the same way — cannot be excluded by cross-comparison alone, and this paper now contains an example. The 17.49 GHz mode is an artifact of a port representation that both solvers share; the two agree on it to 0.001%, and earlier versions took that agreement, with the f_{LC} anchor, as confirmation of the junction physics. The same versions asserted the absence of a mode that Palace had not been asked for, and they aimed every optimization at an anchor derived with the wrong formula. Agreement with a reference bounds implementation error, not shared modeling choices, the reference's own search window, or the reading of a published number. The brackets are certified for the discrete polygonal geometry, in plain f64 arithmetic; the certificate's code is validated against analytic cases and mutation tests, not formally verified. The bare and loop-corrected targets assume that coupling and loop inductance extrapolated on geode-fem's meshes describe DeviceLayout's run; on the coarse fixture's own values the targets move by up to 0.21 fF. The two refined runs share a mesh family, a marking rule and an element order, and Palace's agreement on the adapted mesh is agreement on the same discretization. The choice of P2 elements and ten refinement cycles was made after exploratory runs and was not preregistered. No two-target or coupling-constrained optimization is demonstrated. The parallel-plate tier's one-step convergence is an affine-map property, disclosed as such. The real-device chain uses the scalar trace-averaged sapphire, and the scalar-versus-tensor difference is not converged: it changes sign under refinement (Section 7). Repeat counts and timing boundaries differ across experiments and are stated in each table. The large-scale eigenmode completion used

a different 128 GB box and lacks mode-content verification; it cannot establish equivalent-work throughput.

Limitations. (i) One geometry. The eigenmode results are on one mesh at first order; higher-order and refined eigensolves are future work (a $p = 2$ Palace configuration is committed for that purpose, and the coupling and loop-inductance studies ran Palace at $h/2$ and at second order for those two quantities only). The electrostatic branch has a convergence study, and it shows the committed mesh far from converged for C_Σ (Section 7). (ii) The lossless model leaves readout ports open; resistive terminations (hence complex-symmetric pencils and mode Qs) are out of scope. (iii) The credential eigenpath remains the ungauged committed benchmark with the default port term, which retains the 17.49 GHz port-sheet artifact, and mode classification under it still uses the Palace reference spectrum. The rank-1 port-voltage shunt that removes the artifact is built and opt-in; whether it becomes the default is open. (iv) The CPU evidence depends on request selection, field output, error estimation and implementation version, and Palace solves about 17% more unknowns from the same input mesh; it supports no unconditional per-core or scaling advantage. (v) GPU evidence covers driven-solve correctness and an honest scaling negative in f32 on a cube fixture family; no GPU eigensolve path exists. (vi) The differentiable branch is electrostatic/LOM only (eigenmode/EPR is roadmap) and scalar-ε for the shape chain. Its real-device result is certified for the polygonal geometry of the committed mesh, with the junction's 5.5 fF outside C_Σ, and the tensor bracket is wide ($[72.96, 89.50]$ fF for X^0). The refined optimization runs in native f64 with hand-derived element integrals, outside the Burn tape. (vii) The design target is uncertain at a level the brackets resolve: the coupling and the loop inductance are not mesh-converged.

14 Conclusion

The discrete-adjoint layer supplies solver-derived material and geometry gradients for electrostatic and driven FEM operators, and the stationary capacitance derivative closes the geometry-to-charging-energy loop. On the real device, a finite-difference-validated P2 shape gradient on an adaptively refined mesh reaches the 93.954 fF anchor, the published 4.14 GHz read as a harmonic frequency, in four accepted steps. That result is for the scalar-permittivity electrostatic C_Σ without the junction's 5.5 fF, and its gradient is a hand-derived equilibrium-energy shape derivative solved iteratively in native f64, outside the Burn tape and without a factorization. Held-out refinement and a certified equilibrated-flux lower bound then show what that design is worth: its converged capacitance lies in $[92.75, 93.00]$ fF, more than 1% below its target, and a resolution-corrected design lies in $[93.89, 94.14]$ fF, which contains the anchor. The optimizer converges to whatever target it is given, and a 0.25 fF bracket decides design against target; what is left uncertain is the target. Removing the measured qubit–resonator hybridization and adding a series loop inductance move it to 93.853 and then 93.143 fF, both estimates on unconverged extrapolations; neither design brackets the last, and a re-optimization at it is the next experiment. The convergence study behind these statements is a result of its own: the committed mesh overstates the scalar-permittivity C_Σ by at least 54.69 fF and the tensor-permittivity one by at least 47.18 fF, the converged device lies below the anchor and not above it in both models (in $[77.40, 77.61]$ fF and below 89.50 fF), and the coarse-mesh optimizations that lowered C_Σ to reach an anchor moved the wrong way while passing every self-consistency check. Same-input-mesh agreement with Palace supplies a forward-solver credential that now includes the transmon qubit mode, which earlier versions wrongly reported as absent, and identifies the 17.49 GHz mode as a shared port artifact that an opt-in shunt removes. The updated mode-verified timing and driven AMS experiments delimit the computational cost without claiming a general performance advantage. Eigenmode/EPR derivatives remain future work.

Acknowledgments. geode-fem's development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline: results are traced to committed, human-auditable artifacts, and statements that rest on a scratch probe, on a reading of Palace's source, or on inference are labelled as such.

A The historical optimization figure, and figure provenance

Figure 9 keeps the two earliest optimization tiers in view: the analytic loop proof of Section 9.1 and the single-parameter runs on the coarse device mesh of Section 9.2. Panel (b) is historical. Its anchor shortfall is an artifact of the unconverged coarse mesh (Section 7), and the mesh-validity budget it shows belongs to that fixed-topology map on that mesh; it is not a limit on the refined optimization of Figure 6.

Every plotted data value in this paper is read by the figure scripts in `figures/src/` from a committed artifact. v8 had one exception, a snapshot of the optimized outline of the superseded 89.9 fF design, taken from a mesh the artifact directory does not commit. v9 draws the superseded designs nowhere, and shows the two current optimized islands in Figure 6c–e only by their committed bounding boxes (`design.json`), because their outlines are not committed either; the snapshot is gone. The scripts need Python 3.11 or newer, NumPy and Matplotlib, and no private plotting style.

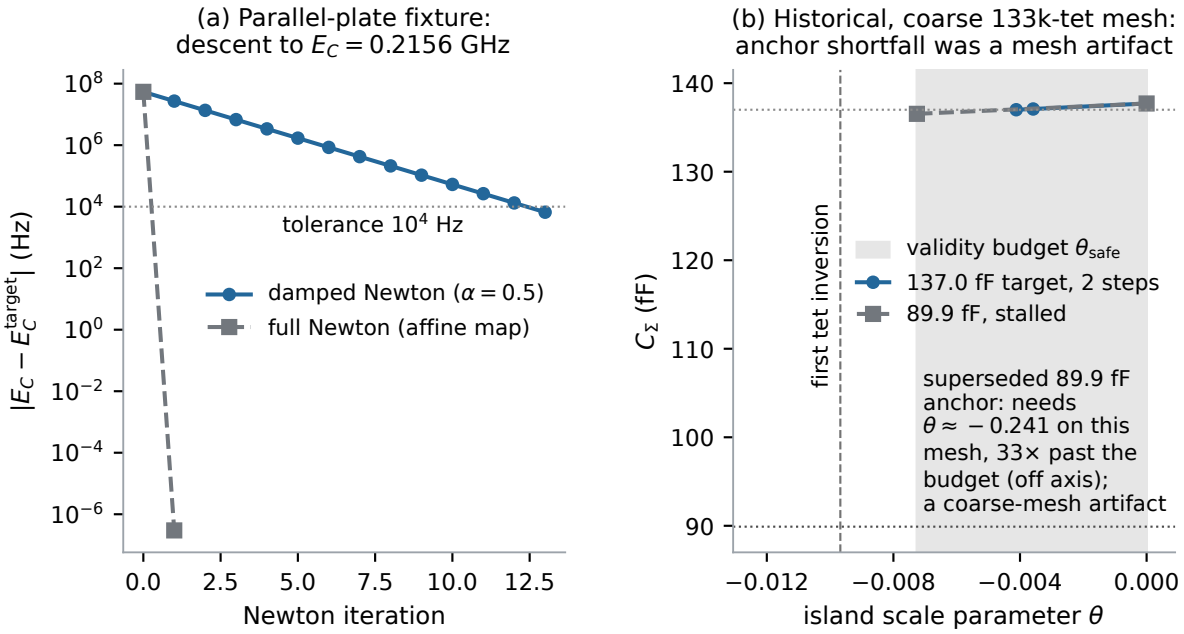


Figure 9: Gradient-based optimization to a target E_C via the discrete-adjoint shape gradient. (a) Parallel-plate loop proof: $|E_C - E_C^{\text{target}}|$ versus Newton iteration; full Newton converges in one step (the parametrization is affine — expected, and disclosed), the damped $\alpha = 0.5$ run supplies the 13-step descent curve; the converged design is confirmed by an independent fresh forward solve at relative error 1.4×10^{-15} (`benchmarks/transmon_diffopt/results.toml`). (b) Historical, on the coarse 133k-tet DeviceLayout mesh under the fixed-topology island-scale parameter θ: a within-budget demonstration target ($C_\Sigma = 137.0$ fF) converges in two genuine (non-affine) Newton steps, fresh-solve confirmed at 5.6×10^{-6}; the bounded run toward the then-current 89.9 fF anchor, since superseded (Section 7), stalls honestly at the mesh validity budget $\theta_{\text{safe}} = -0.0073$ (first tetrahedron inversion at $\theta = -0.0097$), 33× short of the $\theta \approx -0.241$ the anchor requires on that mesh (`benchmarks/transmon_diffopt/pad_results.toml`). The shortfall is an artifact of the unconverged mesh: converged, the device is already below the anchor (Section 7), and Section 9.4 reports the refined optimization. The target of panel (a), $E_C = 0.2156$ GHz, is the superseded anchor value; the loop proof does not depend on it.

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